

Universe24: the meeting of the past with the future, the least computation that shows nature's clicks

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Abstract

The world is the meeting of the past and the future: the meeting is a click, and an observer sees only clicks. One local rule in whole numbers, Rule3, steps every Node of a cubic lattice from its present, its past and its six neighbours by one Euclidean division whose remainder stays at the Node; every run is exactly reversible. The lattice holds only events, split to the six neighbours at every Node: run forward from an emitter they are the future, every path at once; read backward from a click, the past. A click is their meeting at a declared detector, the one non-local element: the wave is the future, the click the present, the track the past. A family enters as one row, its pair and its shape, with as few free numbers as the rule allows. Gravity is a level that slows a Node's clock once and its Links twice; two features were fitted to light's bending and the perihelion, and against the potential Kepler's orbits measure the same forms reach Einstein's only as the matter family's gap closes, which bounds that gap. Lorentz's factor at a body's own light speed and free fall's universality follow unfitted; the magnetic force between moving charges does not. Every formula is fenced as a lattice reading or a click formula. Running no experiment, we derive the two-slit fringe row and, under a declared credit rule, Bell's correlation with the sum $2\sqrt{2}$, where every local credit gives at most 2.

Keywords: integer lattice rule; reversible cellular automaton; measurement as a meeting of past and future; Bell's inequality; weak-field gravity; discrete relativity

1 Introduction

What our world is. This paper starts from one statement and never contradicts it (ASSUMPTION): the world is the meeting of the past and the future; that meeting is a click; and an observer sees only clicks. We give the statement an exact form. The model is a cubic lattice of Nodes on which one local rule in whole numbers acts at every Node in every interval of time. A Node holds a record's present and its past, two integer levels, and the rule makes the record's future from the two and from the six neighbouring Nodes. The lattice holds nothing but events that split: at every Node a level splits by the rule to the six neighbours, and the record is the sum of the split events. Run forward from an emitter, the split events are the future, every path at once. Read backward from a realised click by the same rule, they are the past, and through two slits both pass both gaps. A click is where the two meet at a declared detector. Which meeting is realised is the one thing the model does not determine, and we name it as the one non-local element of the description (Section 5.2). What appears as a wave is the future, the record run forward with every possibility; what appears as a particle is the present, the click; and the track is the past (Section 5.1). We hold this to be the

least computation that shows nature's clicks: the least state, the least non-locality and the least randomness, in bounded integers, locally and reversibly. That is a statement of economy and not of why nature is so, which we do not claim to know.

The rule in one line. Everything in the paper follows from one line of integer arithmetic, Rule3, together with one write and three readings. At every Node, in every interval and for every family of physical values [1, the line; the booking; the primitives]:

Rule3:	$w a_{\text{next}} + r' = \sum_{a \in \{x, y, z\}} R_a (\text{arr}_{+a} + \text{arr}_{-a}) + S a_{\text{now}} - w a_{\text{before}} + r, \quad 0 \leq r' < w,$ $w = 6 \text{ den } \Gamma^2, \quad R_a = 2 \text{ num } p_a^2,$ $S = 12 \text{ den } \Gamma^2 - 12 (\text{den} - \text{num}) p_0^2 - 4 \text{ num } (p_x^2 + p_y^2 + p_z^2);$
the one write:	a held part at a Node gains $\left(\sum_f w_f q_f(\text{Node}) + r \right) \text{div } E$, the remainder r kept there;
the readings:	$D_i = \text{now}_i^2 - \text{next}_i \text{ before}_i, \quad W_i = \text{re}_{\text{now}} \text{ im}_{\text{before}} - \text{im}_{\text{now}} \text{ re}_{\text{before}},$ $F_{ij} = \text{num} (\text{now}_i \text{ before}_j - \text{before}_i \text{ now}_j).$

(1)

In the first line a_{now} , a_{before} and a_{next} are the family's level at the Node now, one interval before and one interval later: the present, the past and the future. $\text{arr}_{\pm a}$ is the arrival through the Port $\pm a$, the neighbour's level. r and r' are the remainder kept at the Node before and after the step. Γ is the Node clock, the pace of an empty Node. $[\text{num}, \text{den}]$ is the family's pair, and $\cos \omega_0 = \text{num}/\text{den}$ is its rotation per interval at wave number zero. p_0 is the Node's clock pace and p_a its pace along the axis a ; both are the Node clock lowered by the field levels the family reads there (Section 3.4). div is the floor division. The second line is the only way a field's level is ever written: the sum over the sourcing families f of each family's own quantity q_f at the Node, weighted by w_f , plus the part's one remainder, divided by the part's wall E . The third line is what is read from a record and never written: its form D_i , which sources the fields; the Wronskian W_i of its two level pairs, its charge; and its current F_{ij} into the Node i from its neighbour j , the conserved flux of Rule3, which a detector reads through its boundary in quanta of $W_c = 3 \text{ den } T$, with T the quantum's action. Everything else is a number in a file.

What the paper contributes. We state the rule and prove what is exact about it on the lattice: every run is reversible to the bit, the rule conserves a quadratic form whose Link term is a current, and the flux of momentum is the wave's own stress (Section 4). We formulate measurement as the meeting of the forward and the backward record at a declared detector, with the record never ending and the one non-local element named (Section 5). We derive the families of physical values as bands of the rule, each entered as one row of a file, and choose a universe with as few free numbers as the rule allows (Section 6). We show that bodies are fixed points of the rule, prove a stability theorem for them, and state how bodies cluster and what each thing represents in a cluster, the nesting of the three levels at a Node, the three kinds of binding by rank and the click as the only thing between clusters (Section 7). We derive the weak-field forms of gravity and Lorentz's factor from the rule's coefficients, mark each by what it rests on, and state them both in the model's own potential and in the potential that nature's tests of gravity read (Section 8). We state what a detector reads of a laid record, of a pair and of a body as formulas of clicks, each with its status (Sections 5 and 7). We state the implementation's two gates, the two slits and Bell's, as derivations of what each must give, without running either (Section 9).

What is put in. We separate what the rule gives from what we chose. Three choices were made with a known result in view, and we state them as inputs, not predictions. First, a field's level enters a Link's pace twice, a choice fixed by the bending of light; the bending and the delay of light

are therefore consistency checks. Second, the clock's pace carries a second-order term, adopted with the perihelion's advance known; the perihelion is therefore a consistency check. Both hold in the clock's own potential; against the potential that Kepler's orbits measure, the same three forms reach Einstein's only as the matter family's gap closes, which bounds that gap and stands against the rule's own matter pair (Section 8). Third, a click's share is the square of the meeting, a credit rule chosen to reach nature's correlations; Bell's value $2\sqrt{2}$ follows from it and does not test it. Five numbers are fixed by nature's measurements: the vacuum content, the binding range, gravity's divisor, the proton's count and the unit of charge. The magnetic force between moving charges does not follow from the rule as it stands, and the paper says what would give it (Section 8, (f)). Table 1 lists every result of the paper with its kind, its fence and its inputs.

How to read the paper. Every claim carries one of six marks. **THEOREM:** proved for every lattice. **DERIVED:** obtained from theorems by argument. **COMPUTED:** a number obtained from the rule's lines before any run. **ASSUMPTION:** a line held by decision and named where it is used. **HYPOTHESIS:** a line outside the model, under its own name, with what would decide it. **EXPERIMENT:** a number the model does not fix, with the experiment named that would fix it, what is measured against which number of the model and with which instrument; the paper runs none. A result that rests on an input chosen with that result known is also marked **FITTED**; it shows consistency, not a prediction. Every formula carries, beside its mark, one of two labels, its fence. **GAMEBOARD:** a reading of the lattice's own numbers that no instrument reports, the levels, the form, the Wronskian, the currents, the paces, the bands and the tensions, valid on the lattice and a diagnostic. **CLICKS:** a formula of what a detector reports, the share Q_R/W_c on the forward orbit (Section 5.2) or a ratio of two such, the only kind that nature can be set beside. A GameBoard formula is a step toward a click formula and never a claim about nature. Only a detector's click is a measurement. A number read off the lattice's state is a diagnostic and is labelled so; it is never compared with nature. No number from a simulation run appears in this paper. The law's document, the implementation, the derivations with their checks and the text were worked out with a large language model used as a tool under the author's direction, and the author reviewed every statement; the Statements and Declarations at the end say how. The paper cites three documents of its archive by the names of their sections: the law [1], the engine [2] and the decisions [3]. Section 2 places the model beside related work, Section 3 states the model, Section 4 its exact properties, Section 5 the measurement, Section 6 the families and the universe, Section 7 the bodies and their clusters, Section 8 the derivations, Section 9 the implementation's two gates, Section 10 the constants and the limits, and Section 11 the implementation.

The method of derivation. Every derivation of the paper follows six steps. It starts from Rule3's line alone, the reader's band at the paces its fields give, the clock once and the Link twice; no structure is written before its band is computed. Its source is a field's static rest, the six Ports' Green's function with a body's well as the one number. Its bridge to nature is a body's clicks and nothing else: a clock is a bound body's rotation reading the content once, a source is a body's well, a detector is a body, and an orbit is a body's chain of clicks. Every observable is a formula of clicks. Every form is re-derived whenever a line changes, and what does not come out is named; two independent derivations precede any line of the law. Every result carries its status and enters the paper as a derivation, never as a run's result [3, algebra first; the paper].

The words. The physical location is a *Node*, and its complete local information is its *NodeState*. Nodes are joined by *Links* through six directional *Ports* per Node, and the lattice of Nodes is the *GameBoard* (Fig. 1). A local change is an *Event*, and there is one local rule, Rule3. A *family* is one kind of physical value with its pair and what it holds. Its *record* is its levels over the GameBoard,

Table 1 The results of the paper, each with its kind, its fence and what it rests on. A result marked FITTED rests on an input chosen with that result known, so it shows consistency and is not a prediction; a calibration is a number taken from a measurement of nature; EXPERIMENT is a number the model does not fix, with the experiment named in the text. The fence says whether a formula is a reading of the lattice (GAMEBOARD) or a formula of what a detector reports (CLICKS).

Result	Kind	Fence	Rests on	Section
Every run is reversible to the bit	theorem	GameBoard	Rule3	4.1
The rule's acts form no product of levels	theorem	GameBoard	Rule3	4.2
A conserved form, its current, and the momentum flux as the wave's stress	theorem	GameBoard	Rule3	4.3
The count is the record's share; the total share is never negative	theorem	GameBoard	Rule3	4.4
Three exact rotations	theorem	GameBoard	Rule3	6.1
Light's speed $1/\sqrt{3}$, its dispersion, and no dispersion along the cube's diagonal	derived	GameBoard	Rule3 for the pair [1, 1]	3.3
A click is a meeting, credited by the shares; no collapse; no signal	assumption, derived	clicks	the meeting and the detector	5
A field's reach from its gap, Yukawa's form	derived	GameBoard	Rule3 for a gapped pair	6.4
The stability of a body	theorem	GameBoard	the body's functional	7.2
Free fall is universal for extended bodies of one family	derived, computed	clicks	the fixed point and the binding range	7.3
A body's clicks: no light from a steady rotation, light at a breathing body's beat, the resonance at $2\omega_b$, the count's drift	derived	clicks	Rule3 at the body's Nodes	7.4
A cluster: the three levels at a Node as the nesting, three kinds of binding by rank, the click as the only thing between clusters, the recursive hierarchy	assumption, derived	GameBoard, clicks	the universe is bound; Rule3	7.6
The gravitational redshift, $z = f(\omega_0)U$	derived	clicks	the clock's read	8
Light's bending $4U_b$, its delay and the perihelion's $6\pi U$, in the clock's potential	fitted	clicks	the Link read twice; the clock's second order	8
The same forms against Kepler's potential: $\gamma_K = \text{den}/\text{num}$, $\beta_K = (\text{den} + \text{num})/(2\text{num})$, $1 + \alpha_K = \tan \omega_0/\omega_0$	derived, against nature at [2, 3]	clicks	the fall's formula	8
Lorentz's factor at a body's own light speed	derived	clicks	the body's band	8
The force between parallel moving charges	not derived under the rule; hypothesis	clicks	a phase on the Links	8
De Broglie's relation, refraction, and Sagnac's effect with the group speed	derived	GameBoard, clicks	the band	8
Newton's pull, the clock's G and Kepler's G	derived	GameBoard, clicks	Rule3 for [1, 1] and E_g	10.1
The two-slit fringe row and the lattice's arrival interval	computed	clicks	Rule3 on the declared world	9.2
Bell's $E = \cos 2(a - b)$ and $S = 2\sqrt{2}$, and $S(\rho)$ for unequal parts	fitted	clicks	the credit rule, the square of the meeting	9.4
Every local credit gives $S \leq 2$	theorem	clicks	Rule3 and a local credit	9.4
The vacuum content, 0	calibration	clicks	GW170817	6.5
The binding range, 20 Links	calibration	GameBoard	the proton's 1,836 quanta	7.3
Gravity's divisor, the proton's count, the unit of charge, the matter pair's gap, the Links' wall	calibration, experiment	clicks	nature's measurements	6.6, 10.3
The proton's size, the neutron's sign, frame dragging 0, the waves' polarisations, the post-Newtonian parameters at [2, 3]	derived, against nature	clicks	Rule3	10.5

two per Node with a remainder. A family of quanta has a *dimension*, 1 for one real line and 2 for a plane of two lines; a part of a record, one level pair with its remainder, is what the implementation

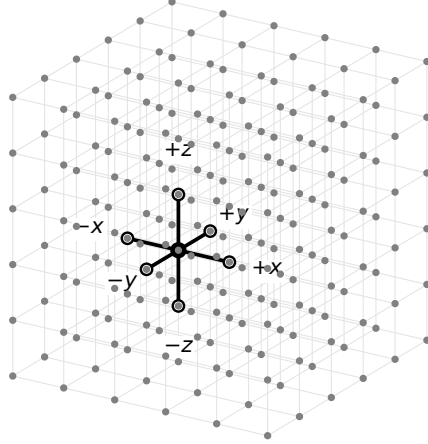


Fig. 1 The GameBoard in space: a cubic array of Nodes, here $6 \times 6 \times 6$, each joined to its six neighbours by Links through the Ports $\pm x$, $\pm y$, $\pm z$ (the faint lines); one Node and its six neighbours marked, the Nodes one interval away from it, whose convex hull is the octahedron of Fig. 2. A Node keeps each family's two levels and its remainders there, and nothing else. Drawn from the definitions; no run.

calls a line. A *body* is the set of Nodes on which a family's record stands bound; its quanta are the record's share there, and it has no law of its own. A *cluster* is bodies standing in one another's wells, bound by the same fields and clicking at one another; a stone is one. A *message* is a travelling record laid by the world, such as an emitter's packet of light. A *detector* is a region of Nodes declared in the file to report, and its *click* is one quantum credited to it from the current it saw through its boundary; the click is the one measurement. An *interval* is one step of the whole GameBoard. A family that holds a value for others to read is a *field family*, or *holder*: the holder of the content slows the clocks of the families that read it, and the holder of the sign carries the charge's field [1, the objects][2, the words].

2 Related work

Reversible cellular automata and digital physics. The idea that the physical world is a discrete computation goes back to Zuse [4]. It was developed as reversible computation and reversible cellular automata by Fredkin, Toffoli and Margolus [5][6][7], as a general programme by Wolfram [8], and as a deterministic foundation beneath quantum theory by 't Hooft [9][10]. Rule3 is a second-order cellular automaton in the sense of Toffoli and Margolus: the next state depends on the present state and the previous one, as $a_{\text{next}} = f(a_{\text{now}}) - a_{\text{before}}$, which is reversible by construction [7]. What the model adds is an integer alphabet with one Euclidean division whose remainder stays at the Node, which keeps the construction exact for rational coefficients, and the reading of the automaton's quadratic forms as physics: the form, the current and the share. Feynman noted that a local classical computer reproduces no correlations beyond Bell's bound [11]; the model agrees, and places the excess in the declared credit of the click rather than in the lattice (Section 9.4).

Lattice schemes and lattice physics. At the vacuum's pace Rule3 is the explicit second-order difference scheme for the wave equation on a cubic lattice, and its guard is the stability condition of Courant, Friedrichs and Lewy [12]: a pace beyond the edge P lets the mode at the zone's corner grow (Section 3.4). Lattice gauge theory uses a lattice as a regulator and takes the continuum limit

[13]; here the lattice is the physics, the Link and the interval are units, and the band’s dispersion is a prediction and not an artefact (Section 10.3). Quantum walks and unitary cellular automata reproduce the Weyl, Dirac and Maxwell equations in their continuum limit [14][15][16]. They take complex amplitudes as the state, whereas here the state is integer levels, the quantum-like features are readings of the record, and the measurement is a separate declared instrument.

Time-symmetric accounts of measurement. The meeting of a forward and a backward description at the point of measurement has a history. Wheeler and Feynman described radiation as an interaction between the source and the absorber, with advanced and retarded fields together [17]. Cramer’s transactional interpretation has an offer wave travelling forward and a confirmation wave travelling backward, meeting at the absorber [18]. The two-state vector formalism of Aharonov, Bergmann and Lebowitz evolves one state forward from the preparation and one backward from the outcome [19][20]. Price argued that a time-symmetric account is what Bell’s theorem leaves open [21], and Wharton and Argaman showed that every locally mediated reformulation of quantum theory must let some mediating parameters depend on measurement settings in their future [22]. The model belongs to this class: the backward reading from a realised click depends on the detector’s setting, and nothing else does. It differs in three ways. Both directions are the same integer line on a lattice that is exact and reversible to the bit. The product of the two directions is a reading, the form D , and not a wave. And the choice among the meetings is named as the one non-local element rather than derived. In the terms of Harrigan and Spekkens the record is an ontic state, a real state of the lattice and not a state of knowledge [23]; what such a state costs in locality is confined here to the credit.

Bell’s theorem and local models. Bell’s inequality bounds every local account of a pair’s correlations [24], in the form that experiments test [25], and Fine showed that a local model, a joint distribution of all the observables and the inequalities are one condition [26]. The model’s local credits give $S = \sqrt{2}$ and $S = 2$, as they must (Section 9.4). Its credit by the contrast reaches the quantum correlation at the efficiency $2/\pi$ by discarding the pairs near the contrast’s zero, which is Pearle’s model of data rejection [27]. The efficiency below which such a model is possible is $2/3$ for a suitably chosen pair [28] and 0.83 for a maximally entangled one [29], and $2/\pi = 0.64$ lies below both, as it must. The loophole-free tests of 2015 [30][31][32] close this road for nature. Superdeterminism, in which the settings share a cause with the pair [10][33], is a different road, which the model does not take: its settings are declared and not produced by the lattice.

Discrete gravity and its tests. Regge calculus [34] and causal sets [35] discretise the geometry of space-time itself. Here the lattice is never curved: gravity is a level that slows the rule’s clock once and its Links twice, and the weak-field forms are read from the coefficients (Section 8). In the parametrised post-Newtonian framework in which the tests of gravity are stated [36], the two fitted choices of Section 3.4 are $\gamma = 1$ and $\beta = 1$. The universality of free fall, which the model derives for extended bodies of one family, holds in nature to 10^{-13} on the ground [36] and to 10^{-15} in orbit [37].

Bodies as fixed points. A body here is a self-trapped mode of the rule, the lattice form of Pekar’s polaron [38]. Its stability theorem is Derrick’s scaling argument [39] with Pohozaev’s identity [40] on the lattice, and the saddle between its two branches is read by the criterion of Vakhitov and Kolokolov [41]. The binding field’s screened reach is Yukawa’s [42].

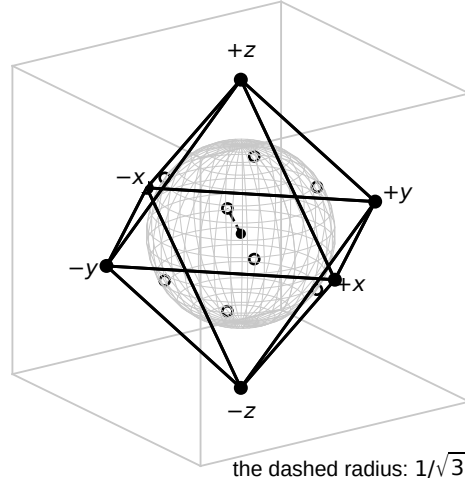


Fig. 2 The six Ports of one Node, $\pm x$, $\pm y$, $\pm z$, and the octahedron they span: the Nodes one interval away. Its inscribed sphere, of radius $1/\sqrt{3}$ (the dashed radius), touches the eight faces on the cube's diagonals (the open circles), and that radius is light's speed at long wavelength, $1/\sqrt{3}$ Links per interval [1, the constants]. The maps of the six Ports onto themselves that keep opposite Ports opposite are the 48 signed permutations of the axes, the symmetry group of the cube drawn faint behind it, under which the law is covariant. Drawn from the definitions; no run.

3 The model

This section states the model: the lattice, the families, the rule, the paces through which one family acts on another, the one write, the readings and the order of acts in an interval [1, the GameBoard and the families; Rule3; the primitives].

3.1 The GameBoard

The GameBoard is a box of $N_x \times N_y \times N_z$ Nodes. Two Nodes that differ by one step along one axis are joined by a Link, and each Node has six Links, one through each of its six Ports, $\pm x$, $\pm y$ and $\pm z$ (Fig. 2). Each axis is declared periodic, open or folded. Beyond an open face no Node lies and a level there is read as 0; a folded axis has one layer, and its two Ports return the Node itself. An inner face is a plane of Nodes declared closed except for declared gaps, which is how a wall with slits is made. A face may also be declared receding (ASSUMPTION): the box then grows by a layer of zeros whenever a nonzero level reaches the Node next to the face, so no event ever meets the face, and a run on such a box is a run on the unbounded lattice. On a receding face the fields build their rest behind the front by Rule3 itself, as Huygens' construction on the lattice, and nothing reflects; the box grows and nothing stretches, so a receding face gives no cosmological redshift [1, the objects; the unbounded board][3, the receding face].

The maps of the six Ports onto themselves that keep opposite Ports opposite are the 48 signed permutations of the axes, the symmetry group of the cube. The rule is covariant under them and under the translations of the lattice. No boost is among them, so Lorentz's group is not a symmetry of the lattice; Section 8 shows how Lorentz's factor nonetheless appears in the clicks.

The NodeState holds the model's own numbers and nothing else: for every family and every part, two levels and one remainder, and for every part of a field family, the remainder of its one write. A count, a current, a charge or a tension is a reading computed from these numbers when it is needed; no register, table, formula or random draw is kept at a Node [1, the objects][3, nothing at

a Node but the law's numbers].

3.2 Families and records

A family is declared by two keys: its pair [num, den] and its shape. A family of quanta, such as matter or light, declares its dimension, 1 for one real line or 2 for a plane of two lines, and holds nothing. A field family declares its sources, the form, the form with the three tensions, or the Wronskian, which fix its parts, one real line per source, and with them its divisor E_s and, for the massless field, its rest; a field sourced by the form or the tensions holds the content, which slows the clocks of the families that read it, and a field sourced by the Wronskian holds the sign, which carries the charge's field. A family's record is its levels over the GameBoard: at every Node a level now and a level one interval before, with a remainder. A family of dimension 1 lays its record as one real level pair, and a family of dimension 2 as a plane of two level pairs (re, im), two solutions of the same line stepped by one call of the rule; only a plane carries a charge, and a neutral body and a charged body of the same pair are two families, two rows of the file with the dimensions 1 and 2 (Section 5.5). A body is a connected set of Nodes on which a family's record stands bound (Section 7), and a message is a travelling record laid by the world [1, the objects].

3.3 Rule3

At every Node, in every interval, every family steps by the first line of Eq. (1). The six arrivals through the Ports, each weighted by its axis's coefficient R_a , the Node's own level times S , minus w times the level before, plus the remainder the Node kept, are summed as one whole number. That whole is divided once by the wall w . The quotient is the next level, and the remainder, $0 \leq r' < w$, stays at the Node and never travels. The whole is taken apart over the Ports and put together again at the neighbours, so nothing is rounded away and nothing is lost; the step back is the same division, which is why every run is exact and reversible to the bit (Theorem 1) [1, the line].

The name comes from the three axes. The Link's paces (p_x, p_y, p_z) are a vector of three directions, the line has one coefficient per direction serving that direction's two Ports, and the six Ports and the wall's factor 6 are the three directions twice. In time the rule is a three-level recurrence, before, now and next. For light the three appears plainly: the next level is the sum of the six arrivals over 3, minus the level before.

At the vacuum's pace, $p_0 = p_a = \Gamma$, the self coefficient vanishes and the line is the second-order wave rule of the pair,

$$a_{\text{next}} + a_{\text{before}} = \frac{\text{num}}{3 \text{ den}} \sum_a (\text{arr}_{+a} + \text{arr}_{-a}), \quad (2)$$

whose rotation at the wave vector $\mathbf{k}$ is $\cos \omega = \frac{\text{num}}{3 \text{ den}} \sum_a \cos k_a$. At wave number zero $\cos \omega_0 = \text{num}/\text{den}$, and for the massless pair [1, 1] along one axis $\cos \omega = (2 + \cos k)/3$, so $\omega = |k|/\sqrt{3}$ as $k \rightarrow 0$ (DERIVED, GAMEBOARD). Light's speed at long wavelength is therefore $1/\sqrt{3}$ Links per interval, below the causal bound of one Link per interval, and its group speed falls at shorter wavelengths as $1 - 0.083 k^2$ [1, the lattice constants]. In the implementation the line is one function, `core/rule3.py`, called on whole-board arrays with the direction +1 or -1, and every level that crosses a Link crosses it through `core/ports.py` [2, the gates].

3.4 The paces: how a field acts on a record

A field acts on a family by changing the family's paces at a Node, the coefficients of Rule3 there:

$$p_0^2 = (\Gamma - c)^2 + c^2, \quad p_a = \Gamma - 2c - t_a, \quad c = \sum_{\text{reads}} \text{weight} \times \text{the read family's time level}, \quad (3)$$

Here c is the content the family reads at the Node, the sum over its reads of the weight times the read field's level, and t_a is the axis content from the fields' tensions on the axis a . A positive weight makes a hollow, an attraction; a negative weight makes a hill. A family with no reads steps at $p_0 = p_a = \Gamma$. The pace is the clock's slowing: a record at a Node of pace p rotates and moves as a record at Γ does, with its intervals p/Γ as long. Every family of quanta reads every holder of the content at the weight 1, so $x = (L_b + L_g)/\Gamma$ with L_b the binding field's level and L_g gravity's; a family of dimension 2 also reads the holder of the sign plainly, and light, the holder's own record, reads none; and a family sources what it reads at the weight it reads with, so every pair of families that read each other is reciprocal [1, the paces; the reads are the held row's pair's]. In the implementation the paces are read in **features/read**.

Two choices fixed with a known result. The level enters the clock's pace once and each Link's pace twice. Nothing in Rule3 forces the factor two on the Link: it was chosen so that light passing a body bends by twice the clock's share, as nature's light does [36] (ASSUMPTION, FITTED). The clock's pace carries the second-order term of a slowing composed unit by unit, $p_0 = \Gamma(1 - 1/\Gamma)^c$, whose square to second order is $(\Gamma - c)^2 + c^2 = \Gamma^2(1 - 2\Phi + 2\Phi^2)$ with $\Phi = c/\Gamma$. This term was adopted with the perihelion's advance known: the clock to first order gives 7π times the potential per orbit, and the clock to second order gives Einstein's 6π (ASSUMPTION, FITTED) [1, the clock once and the Link twice; the clock's second order is the composition][3, the Node clock and the ruler]. In the post-Newtonian language of tests of gravity the two choices are the parameters $\gamma = 1$ and $\beta = 1$ [36] (DERIVED). They are inputs of the model, and Section 8 says where each is used. Rule3's coefficients take the paces' squares as written, with no rounding and no square root.

The band at a pace. From the coefficients, the rotation of a record at a Node of clock pace p_0 and Link paces p_a is (DERIVED, GAMEBOARD)

$$\cos \omega = 1 - \left(1 - \frac{\text{num}}{\text{den}}\right) \left(\frac{p_0}{\Gamma}\right)^2 - \frac{\text{num}}{3 \text{den}} \sum_a \left(\frac{p_a}{\Gamma}\right)^2 (1 - \cos k_a), \quad (4)$$

The mass term scales with the clock's square and the band term with the Link's. Two readings follow. First, the clocks shift alike: at $k = 0$ every massive band's rest rotation obeys $1 - \cos \omega_b = (p_0/\Gamma)^2(1 - \cos \omega_0)$, one factor for every family, so two bound bodies at one Node shift their clocks alike. Second, along a Link every wave's speed falls with $p_a = \Gamma - 2c - t_a$, so light is delayed and bent by twice the clock's share. The line is not conformal: a light wave at a fixed wave number scales with $(p_a/\Gamma)^2$ and a bound mode at rest with $(p_0/\Gamma)^2$. At a Node of content $x = L/\Gamma$ light's band is $[2 - 4(1 - 2x)^2, 2]$ in $2 \cos \omega$, and at long wavelength $\omega = (1 - 2x)|k|/\sqrt{3}$: light's speed is $(1 - 2x)$ times its vacuum speed, an index $n = 1/(1 - 2x)$ with $n - 1 = 2x$ (Fig. 3). A record whose rotation lies outside the band at a Node is evanescent there and reflected whole; light meets a total mirror where $p_a < \Gamma \sin(k/2)$ [1, the band at a pace; the rows against nature, (b)].

The guard. A pace must be positive and below an edge P , with $P^2 = 2 \text{den} \Gamma^2 \text{div} (\text{den} + \text{num})$, which is Γ^2 for light. Beyond P the mode at wave number π on every axis has the factor $2 - 2(1 - \text{num}/\text{den})(p_0/\Gamma)^2 - 4(\text{num}/\text{den})(p_a/\Gamma)^2$ below -2 and grows without bound; at or below P it stays at or above -2 (THEOREM, GAMEBOARD). The guard is checked once, on squares,

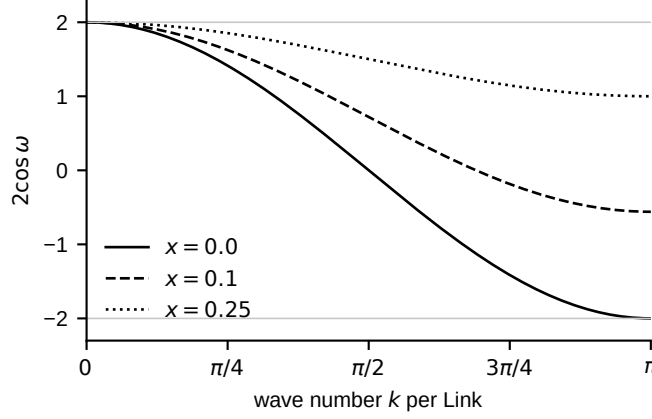


Fig. 3 Light's band at a content $x = L/\Gamma$ read into the paces (DERIVED, Eq. (4) at num = den; GAMEBOARD): the doubled cosine of the rotation along one axis against the wave number, $2 \cos \omega = 2 - 2(1 - 2x)^2(1 - \cos k)$, at $x = 0, 0.1$ and 0.25 . The band's bottom rises with the content, its top stays at 2; at the horizon $x = 1/2$ the band closes to a point. The slope at $k = 0$ is light's speed, $(1 - 2x)/\sqrt{3}$ Links per interval. Drawn from the definitions; no run.

over the whole initial state, and never during a run; in a run no act reads a Node in order to refuse. At $R_a = 0$, a content of $\Gamma/2$, a Link closes: that is a horizon. The clock never stops there: $(p_0/\Gamma)^2 = (1 - \Phi)^2 + \Phi^2 \geq 1/2$ for every content, with its minimum at $\Phi = 1/2$, so at the horizon the Links close while the clock runs at $1/\sqrt{2}$ of the vacuum's rate, a Link horizon with a running clock, and the rotation of a Node whose three Link paces are 0 is $2 \cos \omega = 1 + \cos \omega_0$ from the self coefficient alone (DERIVED, GAMEBOARD). Such a frozen Node is cut from its six neighbours: no current crosses its Links, its share is not a number and no number is invented for it, and it stands in no instrument's front, so the horizon is dark [1, the frozen Node has no share]. At a negative content light's zone-corner mode grows, so light refuses a content below 0, and the vacuum's content is 0 (Section 6.5) [1, the guard].

3.5 The write, the readings and the interval

The one write. Every write into a field's level is one act, the second line of Eq. (1): at a Node in each interval a held part gains $(\sum_f w_f q_f + r) \text{ div } E$, one division per part with one remainder r carried at that Node. Here q_f is the sourcing family's own quantity at the Node: its form D_f for a holder of the content, its Wronskian W_f for the holder of the sign, or its tension $T_{aa,f}$ for an axis part. w_f is the weight with which that family reads the written field, and E is the part's wall, $E_s T$ for the time part and $E_s W_c$ for an axis part. The sources are summed in the numerator and never divided apart, so no remainder stands per sourcing family [1, the primitives][3, the one algebra]. In the implementation this is `features/hold`.

The readings. A reading, which we also call a booking, is a bilinear form of the levels with declared coefficients; it writes nothing on the GameBoard. The law reads three, the third line of Eq. (1). The form $D_i = \text{now}_i^2 - \text{next}_i \text{before}_i$, the present against the product of the future and the past, is $B^2 \sin^2 \omega_b$ at every interval for a standing record of peak B ; its quotient over T , the well, sources the fields. The Wronskian W_i is the charge (Section 5.5). The current $F_{ij} = \text{num}(\text{now}_i \text{before}_j - \text{before}_i \text{now}_j)$ into the Node i from its neighbour j is antisymmetric and is the Link term of the conserved form (Theorem 3); a wave travelling from i to j has $F_{ij} < 0$. A

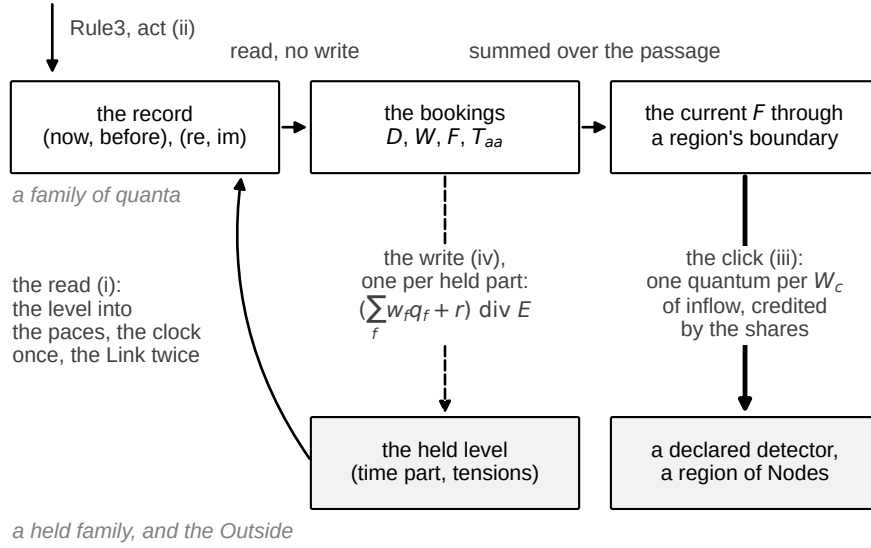


Fig. 4 The three channels through which a family acts on another, drawn from the interval's four acts (Section 3.5). A family of quanta steps its record by Rule3 and books its form and its current; a field family carries a level. The read (i) takes the field's level into the record's paces; the write (iv) takes the record's form, its Wronskian's quanta and its tension into the field's level over the field's divisor; the click (iii) is a declared region's reading of the current through its boundary, summed over the passage, one quantum per wall W_c of inflow, credited by the instrument; the write (iv) is one division per held part of the summed sources. Every arrow is one line of Eq. (1). Drawn from the definitions; no run.

static record has now = next = before, so its form and its current vanish: a static field carries no quanta and no detector sees it, by the arithmetic and not by a clause [1, the booking; the conserved form].

The interval. Every read is of the values at the interval's start; one Link per interval, and nothing in zero time. The interval has four acts in order: (i) the read, each family's paces from the field levels; (ii) Rule3 on every record; (iii) the readings, including each record's six currents at the levels the step started from, its tension, form and Wronskian, and each detector's report, the net current into its region through the region's front boundary; (iv) the write of each field from its summed sources. Light is born at (iv), when a rotating body's varying Wronskian is written into the holder of the sign (Section 5.6) [1, the interval][2, the interval].

The three channels. One family acts on another in three ways and no fourth (Fig. 4). *The read*, at (i), takes a field's level times the read's weight into the reader's paces, once for the clock and twice for the Link; it is multiplicative and is the whole of gravity, of binding and of the charge's pull. *The write*, at (iv), takes the writing family's own quantity into the field's level over the field's divisor; it is additive and is the source of every field and the birth of light. *The click*, at (iii), is a declared detector's reading of the current through its boundary; it is the measurement and writes nothing on the GameBoard. No product of two levels is written anywhere, and no bilinear coupling between families is in the law [3, the coupling is the click alone; light is born by the write].

4 Exact properties

This section proves what holds exactly on every lattice: the reversibility of the rule, the kind of function its acts can form, the conserved form with its current and its momentum flux, and the count as a reading of the record.

4.1 Reversibility

Let σ be the direction, +1 forward and -1 backward; let div and mod be the floor division and its remainder in $[0, w)$; and let ρ be the remainder carried in. With

$$u = \sigma \left(\sum_a R_a \text{arr}_a + Sx - wy \right) + \rho, \quad z = \sigma(u \text{ div } w), \quad \rho' = u \text{ mod } w, \quad (5)$$

the forward call with $(x, y, \rho) = (a_{\text{now}}, a_{\text{before}}, r)$ gives $(z, \rho') = (a_{\text{next}}, r')$, and the backward call with $(x, y, \rho) = (a_{\text{now}}, a_{\text{next}}, r')$ gives $(z, \rho') = (a_{\text{before}}, r)$.

Theorem 1 (the direction). *The backward call returns (a_{before}, r) bit for bit; the reversibility of every step needs no second function.*

Proof. Forward, $\sigma = 1$, the line is the definition of div and mod . Backward, $\sigma = -1$: write Y for the right side's first two terms, $\sum_a R_a \text{arr}_a + S a_{\text{now}}$. Then $u = w a_{\text{next}} - Y + r'$ and $z = -\lfloor u/w \rfloor = \lceil (Y - w a_{\text{next}} - r')/w \rceil$. From the forward line, $Y - w a_{\text{next}} - r' = w a_{\text{before}} - r$ with $0 \leq r < w$, whose ceiling divided by w is a_{before} exactly; and $\rho' = u - w \lfloor u/w \rfloor = w a_{\text{before}} - (Y - w a_{\text{next}} - r') = r$. Checked bit for bit on 20,000 random cases against a forward and a backward function written separately [1, the direction]. $\square$

The whole run therefore goes back: N intervals forward and N back return every array of every family bit for bit, provided every number an act read at an interval's start is one the state after the interval still holds (THEOREM, by construction from Theorem 1). The implementation checks this as a gate: it runs a world forward and back and compares every array at every interval, reporting MATCH or naming the first interval and Node that differ, and a world is not run for results before the gate says MATCH. The condition rules out one kind of read: a coefficient read from a Node's own state across a step cannot be recovered at the step back. This is why a record's rotation sense is never read at a Node, and why the holder of the sign is read according to the record's shape, a plane or a line, and not according to a sense (Section 5.5) [1, the interval][3, the board goes back in time, as a gate; there is no sense].

4.2 What the acts can form

Rule3's line is one act, and declared coefficients make it four: (a) the read, the line with its six arrivals; (b) the one-Node step, the line with the arrivals declared 0, which rotates a level by Chebyshev's recurrence; (c) the division, the line with $R = S = 0$, Euclid's division with the remainder kept; and (d) the load, a declared integer in the numerator, which is how a count enters a level [1, the four acts].

Theorem 2 (what the acts can form). *Every composition of the four acts is a piecewise-linear function of the levels with integer slopes: a sum of linear forms and of floors of linear forms, composed. No such function is quadratic in the levels.*

Proof. Each act is a linear form of the levels or the floor of one; a floor of a linear form is piecewise linear with integer slopes; sums and compositions of such functions are of the same kind. The second difference of such a function in any one level is 0 almost everywhere, while the second difference of a product $\text{now}_i \text{before}_j$ in the pair $(\text{now}_i, \text{before}_j)$ is 1. So nothing bilinear comes from the acts alone. $\square$

So every product of two levels is a reading and never a write, and the three readings of Section 3.5 are the only products the law takes. A line enters the law only if it passes three tests: it is generic (one primitive with declared integers, no family's name), vector (an act on the whole state, with no root and no floating-point number) and local (its own record and the six neighbours, nothing kept at a Node beyond the law's numbers). A line that needs more is a hypothesis under its own name. No act of the interval takes a square root; the integer square root appears only when a world is laid, as the fixed point of the division $\text{act } x \leftarrow (x + n \text{ div } x) \text{ div } 2$ [1, the three tests of every rule][3, the six verbs; the root leaves everywhere].

4.3 The conserved form and the momentum flux

The read matrix $2 \text{ num } p_i^2$ is not symmetric where the paces differ, so the conserved form carries the weight $1/p_i^2$ on its Node terms and the plain current on its Links:

$$E = \sum_i \frac{w (\text{now}_i^2 + \text{before}_i^2) - S_i \text{now}_i \text{before}_i}{p_i^2} - 2 \text{ num } \sum_i \text{now}_i S_6(\text{before})_i. \quad (6)$$

Theorem 3 (the conserved form). *At constant paces E is exact under the line without the rounding; with the rounding each Node's remainder adds at most one unit of the level per interval to the walk of E . Its Link term, the plain current $\text{num}(\text{now}_i \text{before}_j - \text{before}_i \text{now}_j)$ through the Link ij , is the current of Rule3's own conservation law.*

Proof. Without the rounding the line is $\mathbf{a}_{\text{next}} + \mathbf{a}_{\text{before}} = \mathbf{M} \mathbf{a}_{\text{now}}$ with $\mathbf{M}$ the matrix whose diagonal is S_i/w and whose entries on the six Links are R/w . Let $\mathbf{D}$ be the diagonal matrix of the weights $1/p_i^2$; then $\mathbf{DM}$ is symmetric, since $R_a = 2 \text{ num } p_a^2$ at the reading Node and the weight $1/p_i^2$ cancels it on both sides of every Link. For a symmetric $\mathbf{K} = \mathbf{DM}$ the quantity $Q(\mathbf{x}, \mathbf{y}) = \mathbf{x} \cdot \mathbf{D} \mathbf{x} + \mathbf{y} \cdot \mathbf{D} \mathbf{y} - \mathbf{x} \cdot \mathbf{K} \mathbf{y}$ satisfies $Q(\mathbf{a}_{\text{next}}, \mathbf{a}_{\text{now}}) = Q(\mathbf{a}_{\text{now}}, \mathbf{a}_{\text{before}})$: with $\mathbf{a}_{\text{next}} = \mathbf{M} \mathbf{a}_{\text{now}} - \mathbf{a}_{\text{before}}$, $\mathbf{a}_{\text{next}} \cdot \mathbf{D} \mathbf{a}_{\text{next}} - \mathbf{a}_{\text{next}} \cdot \mathbf{K} \mathbf{a}_{\text{now}} = \mathbf{a}_{\text{before}} \cdot \mathbf{D} \mathbf{a}_{\text{before}} - \mathbf{a}_{\text{before}} \cdot \mathbf{D} \mathbf{M} \mathbf{a}_{\text{now}}$, which is the other side. Multiplied by w this is E up to the Link term's sign convention [1, the conserved form]. $\square$

E is a diagnostic. Where the paces move, the total form $D = |\mathbf{a}_{\text{now}}|^2 - \langle \mathbf{a}_{\text{next}}, \mathbf{a}_{\text{before}} \rangle$ changes exactly by $D(t+1) - D(t) = \langle \mathbf{a}_{\text{next}}, (\mathbf{M}(t) - \mathbf{M}(t+1)) \mathbf{a}_{\text{now}} \rangle$, which is 0 at fixed paces and negative where a well deepens (THEOREM, by substitution; GAMEBOARD) [1, the conserved form].

The momentum flux is the wave's own stress. The momentum density along the axis a at the Node i is the current per axis, $P_a(i) = (F_{i,i+a} - F_{i,i-a})/\text{num}$, and the line gives exactly, at constant paces (THEOREM, GAMEBOARD),

$$w [P_a(t+1) - P_a(t)] = \sum_b R_b [G_{ab}(i-b) - G_{ab}(i)], \quad (7)$$

$$G_{ab}(i) = \text{now}_i (\text{now}_{i+a+b} - \text{now}_{i-a+b}) - \text{now}_{i+b} (\text{now}_{i+a} - \text{now}_{i-a}),$$

with $G_{ab}(i)$ the stress reading on the b -Link $(i, i+b)$, the levels now alone. The proof substitutes $w \text{ now}' = \sum_b R_b (\text{now}_{i+b} + \text{now}_{i-b}) + S \text{ now}_i - w \text{ before}_i$ into $w P_a(t+1)$: the S term cancels, the w

term is $w P_a(t)$, and the R term is the difference of the fluxes. The tension on the axis a at a Node, the part a field reads into the Link's pace and the source of a field's axis part, is the momentum flux averaged over the Node's two a -Links,

$$\begin{aligned} T_{aa}(i) &= -\text{num} (G_{aa}(i) + G_{aa}(i - a)) \text{ div } 2, \\ G_{aa}(i) &= \text{now}_i (\text{now}_{i+2a} - \text{now}_i) - \text{now}_{i+a} (\text{now}_{i+a} - \text{now}_{i-a}), \end{aligned} \quad (8)$$

and it is local as F is. A plane wave of amplitude b at the wave number k has $G_{ab} = -2 b^2 \sin k_a \sin k_b$, so the flux along the wave is positive, as a wave's pressure is; a packet of count c and velocity $\mathbf{v}$ has $c v_a v_b$, the stress of dust. With the sign as written a wave's stress deepens the content around it, so light gravitates by its pressure too. The off-diagonal parts and the vector part, which Rule3 never reads, are not written [1, the tension; the final unification][3, the tensor part is the record's own stress].

4.4 The count is the record's share

A family's count of quanta is a reading of its record and no line of its own. At every Node the record's share is

$$e_i = 3 \left[w (\text{now}_i^2 + \text{before}_i^2) - S_i \text{now}_i \text{before}_i \right] \text{div} \left(2 \sum_a p_a^2 \right) - \text{num now}_i S_6(\text{before})_i, \quad (9)$$

the conserved form's Node term over the mean of the three Link paces squared, less the plain Link term, read in quanta over W_c as the nearest whole number. Read with the vacuum's coefficients instead of the Node's own paces, the same expression is the vacuum's share, the unit of the tables of Section 7. Nothing is kept at a Node beside the record and the write's remainder: no count, no remainder of a count and no sense. Over one step of Rule3 the share changes by exactly the six currents into the Node, plus a term that vanishes where the three Link paces are equal, plus Rule3's own rounding (THEOREM, GAMEBOARD):

$$\begin{aligned} e_i(\text{next}, \text{now}) - e_i(\text{now}, \text{before}) &= \sum_j F_{ij} + \text{num} (\text{next}_i - \text{before}_i) \sum_a \frac{(3p_a^2 - P^2) \text{arr}_a(\text{now})_i}{P^2} \\ &\quad - \frac{3 (\text{next}_i - \text{before}_i)(r'_i - r_i)}{2P^2}, \end{aligned} \quad (10)$$

with $P^2 = \sum_a p_a^2$, arr_a the two arrivals of the axis a summed, and r, r' Rule3's remainder before and after the step. The identity was checked by hand and by a test of the implementation. So over any region whose three Link paces are equal, the total share changes only by the currents through the region's boundary and the remainder terms (THEOREM, GAMEBOARD) [1, the count is the record's share; the share's change is the currents]. A formulation that moves whole quanta between Nodes by a division of the current, and the sign by a division of the Wronskian's current, would put two more numbers at every Node that no act of Rule3 reads back; the law does not carry it (ASSUMPTION) [3, the count's line and the sense's line leave the law]. Where the paces move, as under a breathing body, the share read at the new paces drifts with no current crossing any Link; the drift's formula is Eq. (13) of Section 7.4, a diagnostic [1, the paces' own change moves the share with no current]. In the implementation the share is one reading, `share.py`.

The signed integers. The levels are signed integers, the currents are signed by their direction and the Wronskian by its sense, and a click is one whole quantum, never negative. The total share of a closed or periodic GameBoard is non-negative for $|\text{num}| \leq \text{den}$ (THEOREM, GAMEBOARD): with

$\mathbf{n}$ and $\mathbf{b}$ the vectors of now and before and S_6 the six-neighbour sum, a symmetric operator whose spectrum lies in $[-6, 6]$, $\sum_i e_i = 3 \text{ den} (|\mathbf{n}|^2 + |\mathbf{b}|^2) - \text{num } \mathbf{n} \cdot S_6 \mathbf{b} \geq 3 (\text{den} - |\text{num}|) (|\mathbf{n}|^2 + |\mathbf{b}|^2) \geq 0$, with equality only for a static uniform level of light, whose share is 0; in a well the Node term carries the weight $3\Gamma^2 / \sum_a p_a^2 \geq 1$ and the Link term does not, so the bound holds there too. One Node's share can be negative, where its six neighbours' before outweigh its own by more than $6 \text{ den}/\text{num}$, which no plane record does and a lay's edge can; a region's share is negative at most by what stands just outside its front, half a level-square per boundary Port in the current's units. Over the whole passage of one forward packet the inflow accumulated at a region is non-negative up to the rounding, since the lay's current is forward at every Link, so the credit by the shares is well defined with no further line; a window cut inside a passage or a record returning through the front can give a negative reading, and the instrument then credits $\max(s_R, 0)$ (DERIVED, CLICKS) [3, the law's integers are signed].

The lattice constants. The massless field's rest around a static source of s quanta per interval at the divisor 1 is $3G(r)s$, with $3G(0) = 0.7582$ at the Node and 0.2582, 0.1286, 0.0826, 0.0608 and 0.0483 at one to five Links, toward $3/(4\pi) = 0.2387$ times $1/r$ far away (COMPUTED, GAMEBOARD). These numbers and light's $1/\sqrt{3}$ belong to the six Ports and are not numbers of nature [1, the lattice constants].

5 Measurement: the click as a meeting

The click is the meeting of the future with the past at a declared detector, and no act of the GameBoard. This section states what is on the lattice, the meeting, the detector, what a click leaves behind, the charge and light, and what locality costs [1, the click; the reading].

5.1 Events, and no waves

On the GameBoard there are events and amplitudes. An event is a level split by Rule3 to the six neighbours; an amplitude is the level Rule3 produces at a Node, the sum of the split events there; and the record is the amplitudes over the GameBoard. In our world there are clicks, the detectors' reports, and nothing else. What nature calls a wave is, on the lattice, an emitter's events that spread, and a detector that measures them or does not; between the two ends there are events and amplitudes and no wave. An emitter is a message declared in the file, or a body whose write varies. In the two slits the emitter's events pass both gaps, one beam that splits into two fans, every path of the future at once. The fringes are where split events from the two gaps meet at the screen with the same sign, and interference is split events meeting with opposite signs. The past is made of the same split events: read backward from the realised click by the same line, Rule3 at -1 , they form the past's tree, which also passes both gaps back to the emitter. A gap closed by an opaque detector leaves the past one gap. What stays on the GameBoard, the amplitudes no click realised, is not in our world, and it changes a click only where it meets another event that is realised (ASSUMPTION) [1, no waves on the event board][3, no waves on the event board].

5.2 The click is the meeting of the future with the past

Assumption 1 (the click is a meeting, the law and not a hypothesis). A record runs on the GameBoard in both directions by the one line: forward from the emitter, the possibilities, the future; backward from the realised click, what happened, the past, Rule3 at -1 . A click is where the two carry the same signal at a declared detector; its share is the product of the two directions

at the region, the form $D = \text{now}^2 - \text{next} \times \text{before}$, the now against the future times the past; which meeting is realised is the draw, the one non-local place. The detector only asks whether the future and the past met at it.

In plain words: a click happens when the past meets the future; the click is the now; the detector only asks whether the future and the past met at it; and a click is created only at their meeting. This is a line of the model and not a hypothesis [3, the click is the meeting of the future with the past].

The click, algebraically. Let X be the state space: at every Node, for every part of every family, one integer level pair and the write remainders, and nothing else. The step S_t is Rule3 over the interval with the paces read from the fields at its start; it is a bijection of X , exact by the division with its remainder, and affine in the levels at fixed paces; its forward orbit is the future and its inverse orbit the past. The steps generate a group with the lattice's translations, the 48 signed permutations, the time reversal and the plane's rational rotation as its symmetries. The readings D , F and W are bilinear forms on X , covariant under the group and never written: D is even under the time reversal, F and W odd, as a current and a sense flip. An instrument R is a declared region with its front Ports and its coefficients, which define a quadratic form Q_R on X : for one line, the inflow through its front Ports accumulated over the window; for a pair, the window's sum of the squared cross-side sum over the parts. A click is the pair (R, t) , the credit of one whole quantum to the instrument R at the interval t , drawn with the probability $s_R / \sum_{R'} s_{R'}$, where $s_R = Q_R(x_t) / W_c$ is the instrument's form on the forward orbit over the quantum's measure of a current. In the meeting's words, the inverse orbit from the credited state coincides with the forward orbit on the realised path, and the share is the bilinear form's value at the coincidence. The click writes nothing to X and is no element of the group: it is the one map in the model that is not a bijection, from the vector of shares to a unit vector, and it is a scalar under the group, so no click names a Node. Every observable is Q_R / W_c on the forward orbit or a ratio of two such. Three precisions: the window is the whole passage of a packet, or the credit floors at 0 (Section 4.4); the credit's interval within the window is the draw's by the inflow's time profile, which the model gives (Section 9.2); and a Node whose three Link paces are 0 stands in no instrument's front and is never clicked, so the horizon is dark (ASSUMPTION for the definition, DERIVED for the precisions; CLICKS) [3, the click's algebraic definition].

What the backward reading is. Along the realised path the backward reading is the same triples of levels, since the GameBoard is reversible, so it brings no new number, carries no count and no signal, and writes nothing. What it does is choose which tree is the past. The reversibility of the lattice is therefore required by the model and is not a convenience of the implementation: without the exact inverse the past could not be read back from a click, and a pair's common origin could not be reached. A Node in the now reads neither the future nor the past; the two meet only at a click, so a Node is read only through a detector.

One form for every click. At the credit the parts of a record are read with the detector's coefficients, each part paired with the same part through the common origin, summed and squared; the share is that density, and the draw is by the shares. The two slits are the case with nothing to pair: light is one real line, and the share is the record's own inflow. Bell's gate is the case through the common origin: a pair born as one event is two trees with one root, the backward reading from the first click passes the root and continues forward on the partner's tree, and the partner's click meets the re-weighted record (Section 9.3). Absorption by a body is the case of resonance, and emission is the same meeting with the roles reversed. In the implementation the joint share is

computed from the forward records alone, by the linearity of the rule: the instrument reads each part's levels at each port, forms the sum over the parts for every pair of ports, takes its density as the joint share and draws once per pair, the first side from its marginal and the partner from the conditional. No backward Rule3 is stepped; the gate of Section 4.1 shows that the backward reading exists to the bit (DERIVED, CLICKS).

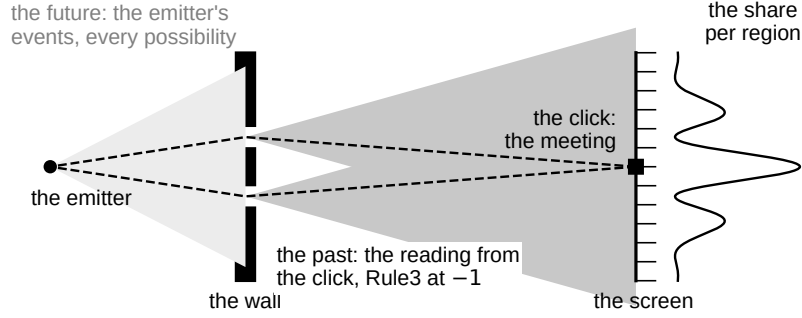
How the meeting looks. Fig. 5 draws the statement schematically. (a) *The future covers every possibility.* The emitter's events split at every Node in every interval; after the wall the record stands at every Node behind both gaps at once, and no path is chosen. One rule at one Node covers all the possibilities by the linearity of the line. (b) *The past is read from the click.* At a declared region of the screen the record's current through the boundary is the share; one wall W_c of inflow is one quantum, and the instrument credits a quantum to one region, the click. From that click the same line runs backward and retraces the realised past through both gaps to the emitter. (c) *Where they meet.* The two directions carry the same signal at the region that clicked and nowhere else; the share there is the product of the two, the form $D = \text{now}^2 - \text{next} \times \text{before}$, and the current F_{ij} is the same product across a Link, the present at one end with the past at the other. (d) *Why two gaps give fringes and a closed gap a hump.* With both gaps open, split events through the two gaps meet at the screen with the same sign where the path difference is a whole wavelength and with opposite signs where it is half of one, so the share along the screen alternates, and the clicks credited by the shares build that pattern quantum by quantum. With an opaque detector in one gap its events leave the GameBoard through the face behind it, the backward reading meets one gap alone, and the share is one gap's envelope (Section 9.2). (e) *The pair.* The click at A's port weights the parts by A's analyser; read back to the root and forward to B, it meets B's analyser's weighting, the root sums the two branches, and the share is the density of the sum, while each side alone reads one half at every port and nothing is signalled. (f) *What an observer sees.* Only the click. Every drawing of the lattice, this one included, is ours from outside the world.

No duality. What is called the particle as a wave is the future: the record run forward by Rule3 with every possibility, which has not yet met the past. What is called the particle as a particle is the present: the click, the meeting, one whole quantum credited at one region in one interval. The particle's track is the past: the chain of realised clicks, and the same record read backward through them. There are not two things but one record read in two directions, and the point where the two readings coincide is what appears as a particle. Between clicks there is no particle, only the future. After a click no wave vanishes; the past has grown by one click. The collapse is not a process but the present: nothing collapses and nothing is erased, the record goes on and the click is a credit. The wave appears as knowledge rather than as a thing because its numbers are real on the lattice and unreadable except at a meeting. A standing body is repeated clicks at one place, and a stone is a past updated every interval (ASSUMPTION) [3, the wave is the future, the click is the present, the track is the past].

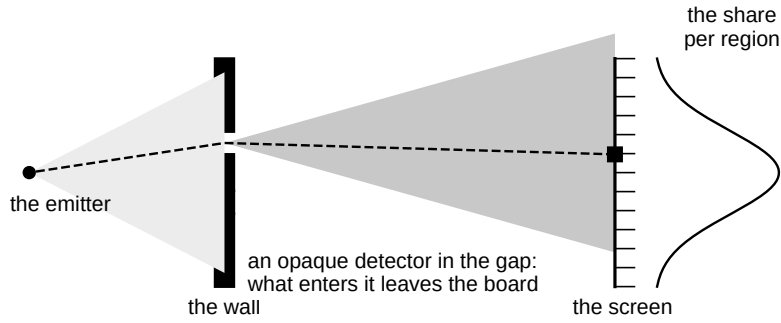
5.3 The detector is a declared instrument

Assumption 2 (the detector's declaration). A detector is one connected region of Nodes declared in the file, never one Node, at least half the wavelength of the family it reports across the beam (q/p Links for the wave $[p, q]$, 4 at $\lambda = 8$); a narrower region, a single Node or a region in pieces is refused by name at load, and the depth along the beam is not gated. It is opaque: backed by a receding face or by the wall's own face, so that what enters it leaves the declared board and never returns, with no act at its Nodes and no log. Its click is its report, per interval and family: the net current into the region through its front boundary Ports, the Ports leading in from the declared board, summed

(a) two gaps: the share alternates



(b) one gap closed: one gap's envelope



(c) a pair: the meeting through the root

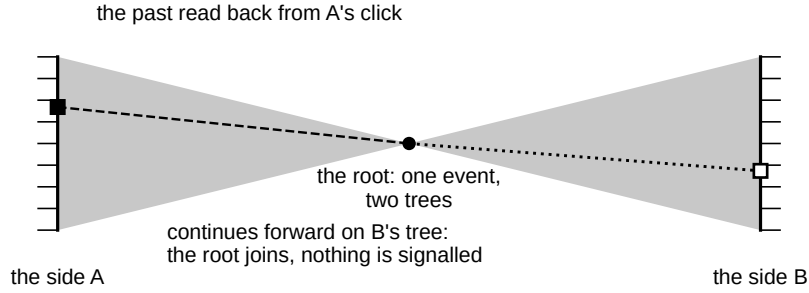


Fig. 5 The click as the meeting of the future with the past, schematically (ASSUMPTION; drawn from the definitions, no run). (a) The two slits: the emitter's events spread forward through both gaps, every possibility at once (the grey fans, the future); from a realised click at a region of the screen the same line is read backward, Rule3 at -1 , through both gaps to the emitter (the dashed lines, the past); the share along the screen, the product of the two directions, alternates. (b) One gap closed by an opaque detector: its events leave the declared board; the backward reading meets one gap and the share is one gap's envelope. (c) A pair, two trees with one root: the backward reading from A's click passes the root and continues forward on B's tree (the dotted line); the root joins and nothing is signalled.

in integers, with the region's name and the family and never a Node; one wall $W_c = 3 \text{ den } T$ of the inflow over a passage is one quantum, so a screen saw N quanta where its regions' inflows sum to

NW_c . The credit: one quantum to one detector, drawn by the shares of the regions' inflows, read outside the GameBoard over every declared detector of the world, one credit per quantum and never two. The credit is the one non-local place of the description and no line of the law.

Why a region: a quantum with a direction has no single Node, since a record confined to one Node contains every wave number at once, $\Delta x \Delta k \geq 1/2$ on the lattice as in nature, and the finest structure a family's amplitudes carry is half its wavelength. So no click names a position finer than a region, and every per-Node number is a diagnostic. Why the net inflow through the front: adjacent regions share transverse Links, and a density that entered one region and moved to the next inside the layer is credited where it entered. Why opaque: a region whose back faces other Nodes passes the density on and reads nothing net, while a region backed by a receding face removes what entered with no echo [1, the click is the detector's report; no click names a Node][3, no click names a Node].

What the model determines and what the instrument declares are separate. What produces a click is known and deterministic: the meeting, the current of the amplitudes through a region's boundary. What is not known is which of the possibilities becomes the measured click: when one possibility is spread over several detectors, the GameBoard gives the shares, no line of the law gives which detector fires, and the instrument draws. Exclusivity, one click per quantum and never two, and Born's rule, the credit's chance equal to the intensity's share [43], are therefore declarations of the instrument and not findings of the model, for nature or against it, as nature's detector efficiency and dead time are [1, what produces a click is not known]. The uncertainty relation is not a separate rule: it is the detector's three declarations, a region of at least half a wavelength, a credit per interval and one draw per quantum, since a meeting is a region and an interval and never a Node and an instant (ASSUMPTION) [3, the uncertainty relation is not a rule of its own]. What each declaration costs, in numbers: a region of n Nodes reads a fringe of period Λ with the factor $\sin(\pi n/\Lambda)/(n \sin(\pi/\Lambda))$, 0.90 for four rows on a fringe of sixteen and 0 at $\Lambda = n$, so the smallest region the model allows reads every fringe the amplitudes can carry except the finest; one click carries no frequency, a frequency being a click rate over a window of N intervals, read to $\Delta\omega \geq 2\pi/N$; and N quanta drawn by the shares give each region $Np_i \pm \sqrt{Np_i(1-p_i)}$, so a pattern is in the histogram and never in one click (DERIVED, CLICKS). In the implementation a detector is a region declared in the world file, its report is the `click` line of the output (`reports.py`), and the credit by the shares is done by the reader of the output and never by the GameBoard [2, the output].

5.4 The click ends nothing

Assumption 3 (the click ends nothing). The click is a detector's reading, local to its boundary; the record goes on after it and nothing ends anywhere. What a detector absorbed leaves the declared board through the face that backs it; the other possibilities go on splitting and leave through the receding faces, events that were not realised, in no click. The ending of a record at a click by the detector is decided against and is no hypothesis of the law.

This is the model's claim and not a deficiency: perhaps nature is local. No signal passes, since nothing at one detector changes the amplitudes at another. A measurement needs no collapse: the record never ends, the click is a reading, and the backward reading from the click writes nothing [3, the click ends nothing]. A bound body can act as an instrument in the same form: the future is light's forward record at its boundary, the past is the body raised by one quantum read backward to the source, and the meeting is the same frequency, the resonance of the body's rotation with the light's; emission is the same meeting reversed, and a body at rest and symmetric clicks nowhere

(DERIVED, CLICKS; Section 7.4). The body's write back waits on the fixed point with the write [3, a Node of the board is read only through a detector][1, the rows against nature, (y)].

5.5 The sign is the rotation sense

A family of quanta laid as a plane carries its record as two level pairs, (re, im), stepped by one call of Rule3 with one set of coefficients, each with its own remainder; a field's parts stay real. The charge density at a Node is the reading

$$W_i = \text{re}_{\text{now}} \text{im}_{\text{before}} - \text{im}_{\text{now}} \text{re}_{\text{before}}, \quad (11)$$

the Wronskian of the two pairs, which Rule3 conserves exactly with the conserved form's weights $1/p_i^2$ (THEOREM, since **DM** is symmetric; GAMEBOARD). Its local change is the Link current num $(\text{im}_i \text{re}_j - \text{re}_i \text{im}_j)$, and its total over the GameBoard is invariant under any change of the paces, since the bracket $\text{im}_{\text{now}}^\top \mathbf{M} \text{re}_{\text{now}} - \text{re}_{\text{now}}^\top \mathbf{M} \text{im}_{\text{now}}$ vanishes for every symmetric **M** (THEOREM, GAMEBOARD). A real record (im = 0) has $W = 0$, so light, the real record of the holder of the sign, is neutral; a rotating record has W of either sign by its sense [1, the sign is the rotation sense; the sign's count is the record's own].

Who reads the holder of the sign is decided by the family's dimension and by one further rule, and by no name (ASSUMPTION, GAMEBOARD). A family of dimension 2 has a Wronskian and reads the holder of the sign; a family of dimension 1 has $W = 0$ and does not, and a neutral body and a charged body of the same pair are two families, two rows of the file with the same pair and the dimensions 1 and 2; the dimension is a key of the row and never a reading of a Node's state. A field is always real, one line per source. A field does not read its own level, so light does not read the holder of the sign, a static charge does not bend light, and light's own W is no charge. A reader's own sense never enters its coefficients, since a coefficient read from a Node's own state would break the exact inverse (Section 4.1). The holder of the sign therefore acts on its readers by a declared act: the plain read of its level into the paces, which is the law today, or a rotation of the two-part record by the angle of its level over Γ , three integer shears with no remainder, under which a record rotating at $+\omega$ rotates at $\omega + \theta$ and one at $-\omega$ at $|\omega| - \theta$. Under the plain read like and unlike charges are not distinguished; under the rotation they would be, and the rotation is Hypothesis 8, not yet part of the law [3, there is no sense; every family has a dimension][1, the sign holder rotates the two-part record].

The charge is the record's. For a record $(\text{re}, \text{im}) = (A \cos \omega t, eA \sin \omega t)$ the reading is $W = -eA^2 \sin \omega$: the charge is the record's size times the ellipticity e of its rotation, which Rule3 conserves, since the two levels obey one linear line. A circular quantum ($e = 1$) has $|W|/D = 1/\sin \omega$, and a nearly real cloud of 1,836 quanta with $e = 1/1,836$ has the charge of one circular quantum. Nothing in the law quantises e , so a body's charge is continuous, and the equality of the proton's and the electron's charges, which nature fixes to 10^{-21} [44], is an input and not derived (Hypothesis 7) [1, the sign is the rotation sense].

5.6 Light is born by the write

Light is the holder of the sign's own record: one record whose time part is a real level pair, stepped by Rule3 at the paces of its reads, so that light is bent by the content, and written by the one write from the Wronskian of the records that read it. A static rotating body writes a static level, Coulomb's field, and radiates nothing. A breathing or moving body writes a varying source at its Nodes, the field carries it away at one Link per interval, and those spreading events are the light

(DERIVED, CLICKS). Its count is the share of that record, read like every family's. A detector reads light's current through its boundary in quanta of W_c , and the travelling events of the other fields, which in gravity's case nature calls a gravitational wave, enter a detector the same way. A body's quanta that leave it stay its family, and no quantum changes family at a crossing. A body reads the holder of the sign at its own Nodes, so a breathing body reads the light it writes and its pace answers, the radiation reaction, in the same loop [1, light is born by the write][3, light is born by the write].

5.7 The price of locality

Three consequences of the model's locality, stated against nature's clicks [1, what is open, item 4; the rows against nature, (g) and (h)]:

- (i) *Exclusivity and Born's rule are declared.* A quantum spread over several detectors is a set of shares of the current at each; the model gives the shares and the instrument gives the click. A dilute packet, under one quantum per Node, builds its pattern click by click by the declaration.
- (ii) *The which-way measurement.* An opaque detector in one gap absorbs that gap's events, whether or not it is credited, and the screen sees the other gap alone, with no fringes: a closed slit by construction, in nature too. A which-way measurement that marks the path without absorbing it is beyond the instrument the model has.
- (iii) *Bell.* A pair is one laid record, a common cause, and its phase is a shared classical variable inside Bell's hidden variable [24]. Every local credit of that phase gives $|S| \leq 2$ (THEOREM, CLICKS; Section 9.4) [25]. The meeting through the pair's root gives $E = \cos 2(a - b)$ and quantum theory's $S = 2\sqrt{2}$ with every quantum counted, while each side alone reads one half at every port and nothing is signalled; this rests on the credit rule, the square of the meeting, which is a declaration of the model and not a derivation (ASSUMPTION, FITTED; CLICKS). The loophole-free test of 2015 that measured the sum found $S = 2.42 \pm 0.20$ [30] [3, the click is the meeting of the future with the past].

5.8 The postulates

The model rests on seven postulates [1, the postulates]:

1. The world is Nodes and Events. Space is the GameBoard; an Event is a change of a NodeState, a level stepped by Rule3 with its remainder; a click is a detector's report of the currents; nothing else happens.
2. A Node holds bounded local information, its NodeState, fixed in size and read from itself and its six neighbours; nothing beyond them is read, no list grows with the world, and nothing of the past is kept.
3. Consistency is local and causal: an Event first changes its own Node and travels Link by Link, one Link per interval. The one exception is the production of a click, the instrument's credit of one quantum to one detector by the shares over every declared detector, which is the one non-local element and is not derived.
4. The causal speed is one Link per interval; every other speed is a rational number of Links per interval.
5. Every physical calculation is on bounded integers, with no floating-point number, no root and no random draw in the law.
6. A measurement is a detector's click: the net current into a declared region through its front boundary over one interval, reported with the region and the family and never a Node. Only a click is compared with nature.

7. The GameBoard, where no one measures, is deterministic and reversible to the bit; the detectors and their clicks are the only thing claimed to represent nature.

5.9 The formulas of clicks

Every observable of the model is Q_R/W_c on the forward orbit or a ratio of two such (Section 5.2). Table 2 collects the formulas of clicks the model gives, each with its status and the section that derives it; the bodies' formulas rest on the body as the fixed point of its binding field (Section 7), and no row rests on a run [3, the click's algebraic definition; the mathematics of the bound bodies and how it bears on the clicks].

6 The families and the universe

A family is a band of the one rule and not a separate law. Its row in the universe file holds only what the rule and the geometry do not fix: its pair [num,den], with $\cos \omega_0 = \text{num}/\text{den}$, and its shape: for a family of quanta its dimension, for a field its sources with its divisor. A pair written over the GameBoard's one denominator Γ makes a mass one integer m , with $\cos \omega_0 = m/\Gamma$ and the wall $6\Gamma^3$; a reduction to lowest terms is a reading and never an act of the loader, so such families share one resolution of their remainders [1, the families from the rule; the mass is one integer; the wall is one].

How a family enters. A family enters the world as one row of the universe file with two keys and nothing else: its pair, and its shape, which for a family of quanta is its dimension, 1 or 2, and for a field its sources, the form, the form with the tensions, or the Wronskian, with the field's divisor E_s and, for the massless field, its rest. The rest is derived from the two keys and declared by no row. Its rotation at rest is $\cos \omega_0 = \text{num}/\text{den}$, and its band follows from Eq. (4). Its coefficients and its wall, $w = 6 \text{den} \Gamma^2$, follow from Eq. (1). Its parts are its sources' count, one real line per source, so the time part alone except for the massless field, which carries three tensions beside it, and a family of quanta has one line or two (Section 6.3). Its reach is set by its gap (Section 6.4). Its reads follow from its dimension by the rule of Section 3.4, and it sources what it reads at the weight it reads with. Every record clicks, a field's included, since no row declares whether it clicks. A new family is one more row of the file and no code (Section 11). Table 3 gives the rows of the universe, and Table 4 gives the numbers of the world beside them [1, a family's declaration; the families from the rule][2, how to add a family].

6.1 The exact bands, and the third band

The line of one Node at rest, $a_{\text{next}} = 2 \cos \omega_0 a_{\text{now}} - a_{\text{before}}$, is exact in integers only where $2 \cos \omega_0$ is an integer. So the rule carries exactly three exact rotations, $2 \cos \omega_0 = 1, 0$ and -1 , with the periods 6, 4 and 3 and the pairs $[1, 2]$, $[0, 1]$ and $[-1, 2]$ (THEOREM, GAMEBOARD). A record on an exact band at rest in the vacuum leaves no remainder in its rest rotation, so it never clicks and only turns. The exact bands are exact only in the vacuum: at a Node of pace p_0 the band's rest rotation is $2 \cos \omega = 2 - 2(\text{den} - \text{num})(p_0/\Gamma)^2$, an integer at $p_0 = \Gamma$ alone, so inside a well a record on an exact band leaves a remainder and clicks like every family's [1, the exact bands; the exact bands are exact only in the vacuum].

The third band, $[-1, 2]$, is allowed by the rule and not used by the universe. It is the mirror of $[1, 2]$, and a family on it holds no standing body: its read is negative, so its own well pushes its rest rotation up into the vacuum's band and the record leaks. The three colours of the strong interaction

Table 2 The formulas of clicks, each with its status: derived from the rule’s line, computed from it, a declaration of the model, a hypothesis, or a number for experiment to fix. Every row is a formula of what a detector reports, CLICKS, and no row rests on a run. U is the clock’s share and U_K Kepler’s potential (Section 8).

The formula	Status	What the model gives	Section
Born’s rule, the count at a region	derived	$N_R = Q_R/W_c$, the share of the inflow through the region’s front over the window; the pattern is in the histogram, $Np_i \pm \sqrt{Np_i(1-p_i)}$	5.2, 5.3
The dead time	declaration	not derivable: the shares credit a region any whole number of quanta in one interval, and W_c over the inflow’s rate is the mean spacing of the clicks and not a least one	5.3
The two-slit row	derived; computed	the near-field minima at the rows 15 and 33; the screen’s inflow peaks at the interval 68.0 against a continuum wave’s 61.8	9.2
The arrival and its spread	derived; computed	$t = L/v_g$ with $v_g = (\text{num}/(3 \text{ den})) \sin k / \sin \omega$; the click’s duration $\sigma_t = \sqrt{\sigma_x^2 + (\omega'' L \sigma_k / v_g)^2} / v_g$ from the band’s curvature ω'' , 4.1 intervals at $k = \pi/4$ over 34 Links	9.2
Bell’s pairs	derived under the declared credit	$E = \cos 2a \cos 2b + \rho \sin 2a \sin 2b$ and $S = \sqrt{2}(1 + \rho)$, 478/169 at the integer settings for equal parts	9.4
The clock’s click rate	derived	$1 - f(\omega_0)U$ for a bound rotation, $f = 2(1 - \cos \omega_0)/(\omega_0 \sin \omega_0)$, and $1 - U$ its small-gap limit; a light clock $1 - 2U$; Lorentz’s factor at the body’s own light speed	7.4, 8
The bending’s clicks	fitted in U ; derived in U_K	$4U_b$; $(1 + \text{den}/\text{num}) 2GM_K/(c^2 b)$ against Kepler’s potential	8
The absorption line	derived (the read’s side); computed when the fixed point with the write is	parametric at $2\omega_b$, the growth $\epsilon/(4 \sin \omega_b)$ and the width $\epsilon/\sin \omega_b$; the two-mode line at $\omega_j - \omega_i$, its rate to be derived	7.4
Emission	derived	a steady rotation and a uniform motion write no light; a breathing body writes at its beat, at the first order in the breathing’s depth	7.4
The which-way sum	derived	the continuity of the one inflow through the two fronts: a region behind one gap reads that gap’s share and nothing of the fringe	5.7
The uncertainty relation	declaration, with its inequalities	$\Delta y k \geq \pi$ from the region, $\Delta \omega \geq 2\pi/N$ from the interval, $\sqrt{Np(1-p)}$ from the draw	5.3
No signalling	derived	$P(A+) = (\cos^2 a + r^2 \sin^2 a)/(1 + r^2)$, independent of b for parts of any amplitudes	9.4
A body’s count in clicks	derived	the share summed over the body’s Nodes, what a full absorber credits; under a moving well it drifts by Eq. (13)	7.4
Magnetism	not derived under the rule; derived under Hypothesis 8	$(1 + 2\beta^2) F_C$ under the plain read, the wrong sign; F_C/γ under the rotation at the wall den T , $\lambda = 1/c^2$: 0.995 at 0.1c and 0.866 at 0.5c	8

are the three axes of a click’s Ports, a detector’s reading of the click’s axis, with no charge of their

own and no force of their own; whiteness is their balance at rest. So the quarks are not in the law: the white body, the flux tube, the charges $2/3$ and $1/3$ and the generations leave with the third band, and the band $[0, 1]$ is likewise allowed and unused [1, the third and the colours, closed as absent][3, the colours are the three axes].

6.2 The rule's own universe

The universe of the model is the one whose every number is the rule's own: the pairs from the integers 1, 2 and 3 of the three exact rotations and the three axes, every divisor 1 except gravity's, and the remaining free numbers named. Its rows stand in Table 3, as the file `examples/events/rule.json` holds them (DERIVED from the two keys of each row) [1, the families from the rule; the rule's own universe][2, the loader and the files].

The universe's integers are the Node clock $\Gamma = 6,000$, a multiple of 6 so that every pair is exact over it; the quantum's action $T = 2^{15}$; and the integers' width, 63 bits, a bound and not physics. Γ and T are units, as c and $\hbar$ are in nature, and no line of Rule3 fixes T , since the rule is linear and the scale of its form is free. Matter reads gravity and the binding field plainly, $x = (L_b + L_g)/\Gamma$, and reads the bound charge; charged matter, a second row of the matter pair at the dimension 2, reads the holder of the sign as well; light reads the same holders of the content and is bent by them, and does not read the holder of the sign, its own field. The bound charge's field is read by neither of the implementation's two gates and leaves the universe file unless a use is found (ASSUMPTION) [3, nothing that is not Rule3 remains]. Every field of the file is a record stepped by Rule3 at its pair and written by the one write; no field is laid by hand during a run, and a world's starting rest is the fixed point of the division act, computed once before the first interval (ASSUMPTION) [1, the final unification][3, two units and the bounds; nothing that is not Rule3 remains].

6.3 What each family carries, and how gravity enters as a clock

What a family carries is set by what Rule3 reads and what a record can write, not by a name (DERIVED, GAMEBOARD). Rule3 reads two things of a Node and no third: the clock's pace p_0 , one number, and the Link's paces (p_x, p_y, p_z) , the three directions of the rule's vector. So a field that acts on the paces carries at most one time part and three axis parts, $1 + 3$. A record of one real line, neutral matter or light, has a form D and a stress and no Wronskian. A record of a plane, charged matter, has a Wronskian W as well. A field is always real, one line per source, since a plane's own W would be its own source. The shapes follow with no choice. *The scalar*, one real line, is the shape of matter and of light, and of the binding field and the bound charge, which are sourced by the form alone. *The plane* is the shape of charged matter alone. *One plus three*, the time part with the three tensions T_{xx} , T_{yy} and T_{zz} , is the shape of gravity, sourced by the form and by the diagonal of the record's own stress (Section 4.3). The pair family of light, for Bell's gate, is *two real lines never summed at a Node*. The law as it stands takes no vector part of a field and no off-diagonal stress, since Rule3 reads them nowhere. A current's read, a part sourced by the momentum density, is a proposed line under check and not the law: for a plane record it can enter only as a phase on the Links, Peierls' coupling, which keeps the band real and the step a bijection, and it is what magnetism needs (Section 8, (f)); for a real record no current's read exists within the three tests, since a read into the Link's pace with the Port's sign makes the band complex and the record grow, a read shifted by the current is a local bijection only when it is uniform, which is a frame, and a swap of a Link's two records drags nothing (DERIVED, GAMEBOARD). Gravity's readers are real records, so the law has no frame dragging of neutral bodies, a finding against nature (Section 10.5); the diagonal stress alone also restricts the polarisations of gravitational waves (Section 10.5) [1, the

Table 3 The rule’s own universe, its five families as the file `examples/events/rule.json` holds them (DERIVED from the two keys of a row, its pair and its shape): each family with its rank, its pair [num, den] (in brackets, written over the Node clock $\Gamma = 6,000$), what it holds at its divisor, its shape (for a field its sources, one real line each; for a family of quanta its dimension), what it reads, and its free number. The pair family of light, two real lines never summed, enters for Bell’s gate in the next version.

Family, its rank	Pair	Holds, at the divisor	Shape	Reads	Its free number
gravity, the time part and the three tensions	[1, 1] ([6000, 6000])	the content, E_g (at least 10^6 at $\Gamma = 6,000$, a click bound from the two slits’ screen; 1,000 in the bodies’ worlds)	1 + 3 real lines, sourced by the form and the three tensions	nothing	E_g , the hierarchy; its rest c_{vac} , 0 by nature’s clicks
the charge, the holder of the sign, whose quanta are light, born by the write	[1, 1]	the sign, 1	1 real line, sourced by the Wronskian of its readers	the holders of the content, at the weight 1; never its own level	none
the binding field, the strong force; its own events have the gap $\cos \omega = \text{num}_b/\text{den}_b$	[2400, 2401]	the content, 1	1 real line, sourced by the form	nothing	the pair, its range $R = 20$ Links, tied to Γ by the proton
the bound charge (the polarisation), the exact band	[1, 2] ([3000, 6000])	the content, 1	1 real line, sourced by the form	nothing	none; a field of under one Link’s reach
matter, the scalar; a quantum that leaves a body is a message of matter	[m, Γ] ([4000, 6000])	nothing	dimension 1, one real line; a charged body of the same pair is a second row of dimension 2, a plane	the holders of the content, $x = (L_b + L_g)/\Gamma$; the plane also the holder of the sign	m , the mass

final unification; the tension][3, every family has a dimension].

How gravity enters: a slowed clock, and no curved space. There is no metric, no force and no law of motion in the model. Gravity enters in three steps. (i) *The write*. At every Node in each interval every family of quanta writes its form over the action, $D_i \text{ div } T$, into gravity’s field over the divisor E_g , by the second line of Eq. (1). A static record writes a static level, a breathing one a varying level. (ii) *The field*. Gravity’s field, [1, 1], steps by Rule3 at the pace 1, so around a static source it rests at Poisson’s solution, $3G(r)s$, which is $3s/(4\pi r)$ far away; a change travels at one Link per interval, retarded, and the exact inverse is kept. The three tensions step beside the time part. (iii) *The read*. Every family reads the field’s level c at its Node into its own coefficients,

Eq. (3): the clock's pace once, $p_0^2 = (\Gamma - c)^2 + c^2$, and each Link's pace twice, $p_a = \Gamma - 2c - t_a$. That is the whole of gravity. A slower clock is what a deeper well is: every massive band's rest rotation slows by the same factor, so two bound bodies at one level shift their clocks alike, a far detector reads the emitter's slow clock, which is the redshift, and bodies of the same family fall alike. What nature calls curved space is here the Link's pace: along every Link through the well a record's speed falls by the level twice, so light's index is $n = 1/(1 - 2x)$. No space is bent. The Nodes stand where they stand and the Links are what they are; only the paces at which the rule divides are lower where the well is deeper, once for the clock and twice for the Link. Of these steps, the factor two on the Link and the clock's second order are the two fitted choices of Section 3.4 [1, the paces; the clock once and the Link twice][3, the Node clock and the ruler].

6.4 The reach of a field, and the binding field

A field's rest solves $\text{num } \Delta^2 a - 6(\text{den} - \text{num}) a = -3 \text{den } \sigma$, its line with the source once an interval, with Δ^2 the six-neighbour second difference. Outside a body its rest therefore falls by $e^{-\kappa}$ per Link with $\cosh \kappa = 3 \text{den}/\text{num} - 2$, the family's own band at zero rotation: the gap is both the mass of the field's quanta and the inverse reach of the field, one number (DERIVED, GAMEBOARD). [1, 1] has no gap, and its rest is Poisson's over the whole GameBoard. [1, 2] has $\cosh \kappa = 4$ and falls by 0.127 per Link, a contact well of 0.41 Links' reach. A pair with num just under den has $\kappa^2 = 6 \text{den}/\text{num} - 6$ to second order, a range $R = 1/\kappa = \sqrt{\text{num}/(6(\text{den} - \text{num}))}$ Links, which is Yukawa's form [42] [1, the well].

Assumption 4 (the binding field and gravity's divisor, decided with a reservation). The well of a record, $D_i \text{ div } T$, is laid into two fields. The binding field, a family with a gapped pair $[\text{num}_b, \text{den}_b]$, $\text{num}_b < \text{den}_b$, at the divisor 1, stepped by Rule3 at its pair as the massless field is and started at its rest, which with the gap is a sum with a range, $R = \sqrt{\text{num}_b/(6(\text{den}_b - \text{num}_b))}$ Links and $e^{-r/R}/r$ beyond it: it binds the bodies, it is the model's strong force; its own travelling events have the gap $\cos \omega = \text{num}_b/\text{den}_b$, a quantum with a mass whose Compton length is R , Yukawa's own line [42] and no input. The massless field [1, 1] at the divisor E_g declared by the file: gravity, long-ranged and weak by E_g . A family reads both levels into its pace at the weight 1.

The first pair is [2400, 2401], with $R = 20$ Links, and the field's quantum has the rest frequency 0.029 per interval. The pair is tied to Γ by the proton: a cloud of 1,836 quanta is bound under $R = 20$ and not under $R = 10$ or 15 (Section 7.3). In worlds with bodies E_g is declared large enough that gravity does not enter a body's binding; 1,000 suffices at the compact pixel. Nature's E_g is about 3×10^{42} , and the hierarchy of gravity and electricity is the file's one number, as G is nature's. The decision carries a reservation, that it may be wrong, and it reopens on any of four findings: a failure of the three tests; bodies that do not stand under the screened sum; a world of two charged clouds that reads the electric over the gravitational pull as $E_g/1,836^2$ the other way from nature; or the reading of scale below [3, the two rows][1, what is open, item 20].

The road not taken. Under every divisor 1 the force that binds a body and the force between bodies are one field, the well read at the weight 1. The gravity between two quanta is then of order one, against nature's 5.9×10^{-39} for the proton, and the electric force between two protons is $1/1,836^2$ of the gravity between them, where nature's is 10^{36} times larger (COMPUTED, CLICKS). *The reading of scale:* matter's rotation at rest, $\cos \omega_0 = 2/3$, fixes the electron's reduced Compton length in the lattice's units at light's speed over the rotation, $0.577/0.841 = 0.69$ Links, while the model's cloud of 1,836 quanta has the radius $2\Gamma/C$, at least 6.5 Links at $\Gamma = 6,000$. Nature's proton is bound thousands of times smaller than the well binds anything, so a single well field binds too

strongly at a distance and too weakly up close to be both the proton's binding and gravity [1, the rows against nature, (v)].

6.5 The vacuum content

Assumption 5 (the vacuum content is 0, a declaration fixed by nature's clicks). The massless field rests at a declared level everywhere, the vacuum content, the key **rest** of the universe file, derived by no line of the law; nature's clicks fix it at 0 in the file's integers at $\Gamma = 6,000$.

A constant is the massless field's own rest, and its step keeps it. Every reader's paces at a rest c_{vac} are $\Gamma - 2c_{\text{vac}}$, the same physics at an effective Γ , so the model cannot tell one rest from another and only nature's clicks can. Light reads the vacuum content into its paces and travels at $(1 - 2x)/\sqrt{3}$, while the massless field's own events travel at $1/\sqrt{3}$, so over L Links the two arrive apart by $2xL\sqrt{3}$ intervals at long wavelength, and by 17.9 intervals over 458 Links at $x = 0.01$ and the wave number $\pi/4$ from the two bands exactly (COMPUTED, CLICKS). Nature's GW170817, a gravitational wave and a gamma-ray burst 1.7 seconds apart after 1.3×10^8 years of travel, bounds the difference of the two speeds between -3×10^{-15} and $+7 \times 10^{-16}$ of light's [45]. So the model's $2x$, gravity's lead over light, is at most 7×10^{-16} , c_{vac}/Γ is below 10^{-15} , and the file's integer is 0: the model's cosmological constant is 0 to the precision of every measurement. At the rest 0 light reads only the bodies' wells. A field with a gap sourced by a changing well rings with its own travelling events, the model's mesons, whose troughs below 0 far from the body light refuses as a real runaway of Rule3 (Section 3.4). The two gates of Section 9 read no rest, since a fringe's spacing and a calibrated phase read none [1, what is open, item 22][3, the vacuum content is 0, a declaration fixed by nature's clicks].

6.6 The world's numbers, and how the universe was chosen

The universe is chosen by one principle stated before any row: one rule closes everything; the families are its inputs and no part of it, each a band of the rule named by its pair; the units are a choice of scale; and the universe holds as few free numbers as the rule allows (ASSUMPTION) [3, the standing goal; not one number in the engine, and as few in the files as the law allows]. The implementation holds no number of physics, so every number of the world is a key of its universe file, and choosing the universe is choosing which keys stand there and why. The road to them:

- (i) *A family is a band of the rule*, and its row holds only its pair and what it holds.
- (ii) *Three exact rotations*: the pairs $[1, 2]$, $[0, 1]$ and $[-1, 2]$ (THEOREM, Section 6.1).
- (iii) *The three of Rule3 is the three axes*: the massless holder of the content carries its time part and three tensions, $1 + 3$, every other family is a scalar, and the parts carry no free number (Section 6.3).
- (iv) *One denominator*: every pair the universe fixes is written over Γ , so a mass is one integer, and Γ is a multiple of 6 so that every such pair is exact.
- (v) *The pairs from 1, 2 and 3* (ASSUMPTION): $[1, 1]$ for the two fields, gravity holding the content and the charge's field holding the sign, whose quanta are light; $[1, 2]$ for the bound charge; and $[2, 3]$, $\cos \omega_0 = 2/3$, the first pair past the exact bands, for matter. The bands $[-1, 2]$ and $[0, 1]$ are allowed and unused (Section 6.1). Nature's tests of gravity read the matter pair's gap: they require den/num within 2×10^{-5} of 1 (Section 8), where $[2, 3]$ is the rule's own choice, so the pair is a number for experiment to fix and the rule's own universe stands against nature there (EXPERIMENT).
- (vi) *Every divisor 1*: with every divisor 1 the universe carried no free number but the mass.

Table 4 The numbers of the world, each a key of the universe file `examples/events/rule.json` and none of the engine: its value in the file, its kind and what fixes it.

The number	In the file	Its kind	What fixes it
the Node clock Γ	6,000	a unit, as c is nature's	a multiple of 6, so that every pair of the universe is exact over it
the quantum's action T	2^{15}	a unit, as $\hbar$ is nature's	declared once; no line of Rule3 fixes it, the rule being linear
the integers' width	63 bits	a bound, no physics	the machine's integers
matter's numerator m	4,000, so $\cos \omega_0 = 2/3$	the mass	the pairs from 1, 2 and 3: [2, 3], the first pair past the exact bands; nature's tests of gravity require it within 2×10^{-5} of light's (Section 8)
the binding field's pair	[2400, 2401]	the strong force, its range $R = 20$ Links	tied to Γ by the proton: a cloud of 1,836 quanta binds under it and not under a shorter range
gravity's divisor E_g	1,000 for the bodies' worlds, 10^6 for the light-only worlds	the hierarchy, as G is nature's	bounded below by light's own fringes, at least 10^6 at $\Gamma = 6,000$ (Section 10.3); nature's about 3×10^{42}
the vacuum content c_{vac}	0	the cosmological constant	nature's clicks, GW170817 (Section 6.5)
every other divisor	1	none	the rule's own universe

- (vii) *Two numbers where nature's scale forced them:* under every divisor 1 gravity is as strong as the binding, where nature's gravity is weaker than the electric force by about 10^{36} ; so the model took two numbers and no more, the binding field's gapped pair and gravity's divisor E_g (ASSUMPTION, with the reservation of Section 6.4).
- (viii) *The vacuum content by clicks:* the one number the model cannot read from inside, fixed at 0 by GW170817 (Section 6.5).

What remains stands in Table 4: two units, Γ and T ; one bound, the integers' width; and four numbers of physics, matter's numerator m , the binding field's pair, gravity's divisor E_g and the vacuum content, with every other divisor 1. The two gates' worlds read a light-only copy of the file with the three holders of the content at the divisor 10^6 , so that light's own well stays under one level. That this universe is the universe is a decision and not a theorem (ASSUMPTION): the rule's integers fix every row, and the model names where another universe could stand. The same integers give a second massive pair, [1, 3], a second mass that the rule does not forbid and the universe does not use, under which nature's spectrum of masses is an open question; and the numbers that would close the universe toward nature, the fine-structure constant, the electron's mass over Planck's and the proton's mass over the electron's, are open and not derived (Section 10.5) [1, the rule's own universe; the final unification; the tree][3, the rule's own universe is the universe; the final unification].

7 Bodies

7.1 What a body is, and how it is computed

A body is its record over its Nodes and the quanta that record carries, and never one Node. Its count, the number of its quanta, is the share of its record (Section 4.4), a reading and not a state. Its form $D_i \text{ div } T$ is written into the fields it sources, and the fields enter Rule3 through the paces

alone. The record of the body's family is bound by the wells its form sources: by the binding field's screened sum with the range R , while gravity's $1/r$ is weak by E_g and enters no body's binding. No law is written for a body. The one rule is written for the Node, and a body is the support of a standing mode of that rule; it has no law of motion of its own, and its record moves by Rule3 alone [1, what a body is][3, what a body is].

Four statements make a body (DERIVED from the line, with the decisions named). (a) *The body is the fixed point of its binding field*: its form sources every field its family sources, its record is the standing record Rule3 makes in those paces, and it stands where the sources return themselves. (b) *The form sources the fields*: a field's level is the rest the start solves from the bodies' forms at the field's divisor, after which Rule3 steps it. (c) *The quanta follow the record*: a body's quanta are its record's share, carried wherever its record's current carries it. (d) *The fall*: a free packet or a cloud accelerates as $a = \Delta |d\omega_b/dc| \nabla c$, with Δ the band's curvature at $k = 0$ and $d\omega_b/dc$ from Rule3's coefficients at the body's pace; a compact body falls by its tail alone, more slowly [1, the four lines of the body].

How a body is computed. By the rule's own iteration and no other formula. Start from a spread of the declared count over the Nodes; write the wells that spread lays into the fields and let them rest; find the record's top mode in those paces by Rule3's own read act, iterated; scale its two levels so that the share carries the declared count; write the wells that record lays; and repeat until the levels repeat, in integers, at the fixed point of the division act. The branch is chosen by the start's width alone: a wide start reaches the extended body, the cloud, and a narrow one the compact pixel, two fixed points at one count, and the iteration never crosses from one to the other. The amplitude unit comes from the integer width and is never written. In the implementation this generator is `tools/pixel.mode.py`, which writes the body's levels into the mode file beside the world; the world file declares only the body's family, its Nodes and their counts, and a check at the start compares the declared count with the share the laid record reads [1, the generator; the four lines of the body][3, the generator is Rule3].

7.2 Stability

Let a body of M quanta have the record ϕ over its region, $\sum \phi_i^2 = 1$, with the counts $c_i = M\phi_i^2$. Let every field r it sources be a hollow ($\sigma_r = +1$) or a hill ($\sigma_r = -1$), with the static kernel $\mathbf{G}_r$ of its line and the divisor E_r , so that the well at a Node is $\Phi_i = \sum_r \sigma_r (M/E_r) \sum_j G_{r,ij} \phi_j^2$. To first order in Φ/Γ the well raises the one-Node rotation $2\cos\omega$ by $4g\Phi/\Gamma$, $g = (\text{den} - \text{num})/\text{den}$, and the body's fixed point is a critical point on the sphere of

$$F[\phi] = \frac{\text{num}}{3\text{den}} \sum_{\text{Links}} (\phi_i - \phi_j)^2 - \frac{2g}{\Gamma} \sum_r \sigma_r \frac{M}{E_r} \sum_{ij} \phi_i^2 G_{r,ij} \phi_j^2, \quad (12)$$

the band's pressure less the wells' energy, each well counted once because the record is its own source (Hartree's form). The record is the top mode of Rule3's symmetric form at the paces, and $dF/d\phi = 2\lambda\phi$ on the sphere is the fixed point's own equation with $2\cos\omega_b = 2\cos\omega_0 - \lambda$ (THEOREM, GAMEBOARD). Let $\phi_R(x) = R^{-d/2}\phi(x/R)$ be the body dilated, with d the GameBoard's dimension; the pressure is K/R^2 and each field's well energy is W_r/R^{s_r} , with s_r the field's scaling power at the body's size: $s_r = d$ beyond the field's reach and $s_r = d - 2$ within it or for the massless field [1, the functional and the dilation].

Theorem 4 (the stable body). *At a fixed point of the binding row of width R the virial $2K = \sum_r \sigma_r s_r W_r$ holds, and the second variation of F along the dilation is $R^2 F''(R) = \sum_r \sigma_r s_r (2 - s_r) W_r$.*

A body is stable only where this sum is positive; it is a minimum of F where further the Hessian on the modes orthogonal to the dilation and the translations is positive.

Proof. $F(R) = K/R^2 - \sum_r \sigma_r W_r R^{-s_r}$; $F'(1) = 0$ is $-2K + \sum_r \sigma_r s_r W_r = 0$; $F''(1) = 6K - \sum_r \sigma_r s_r (s_r + 1) W_r = \sum_r \sigma_r (3s_r - s_r^2 - s_r) W_r$: Derrick's scaling [39], Pohozaev's identity [40] on the GameBoard [1, the functional and the dilation]. $\square$

The factor $s(2-s)$ is +1 at $s = 1$, 0 at $s = 2$ and -3 at $s = 3$: a hollow holds the body where its well falls more slowly than the pressure and breaks it where it falls faster, and a hill does the opposite. Three consequences follow. (a) One hollow beyond its reach binds every mass on a chain, is flat along the dilation on a plane, and in a box has one critical point, a saddle: the body disperses beyond it and collapses within it. (b) A massless field holds the body: Poisson's well in a box has $s = 1$, so $F = a/R^2 - c/R$ has one critical point, $R = 2a/c$, a minimum for every mass, which is Pekar's body [38]. Beside it a hollow beyond its reach adds a barrier, $F = a/R^2 - b/R^3 - c/R$ on a Gaussian body, with the wide critical point a minimum and the narrow one a saddle, and none where $3bc > a^2$, the collapse (COMPUTED, GAMEBOARD). (c) The horizon ends a collapse: beyond first order the well saturates and the Link's pace weakens the pressure, both vanishing together at the pace's floor, so a collapse ends where a Link's pace reaches 0, in a frozen cluster of horizon Nodes [1, the functional and the dilation].

The surplus leaves. At a Node of a body's record the local rotation is read from its three levels, $2 \cos \omega_i = (\text{next}_i + \text{before}_i)/\text{now}_i$, against the local band's top. At a fixed point every Node is bound. A cloud away from its fixed point has a free part, the Nodes whose rotation lies in the band, which leaves at the band's group velocity with its quanta. The bound surplus does not leave by itself, since a bound mode is stationary and Rule3 is reversible: a cloud twice too wide in a static well breathes about its minimum for 6,000 intervals without settling (COMPUTED, GAMEBOARD). Whether a body sheds its excess and cools is Hypothesis 5 [1, the surplus leaves].

7.3 The two branches: the cloud and the compact pixel

The fixed point of the binding field has two branches at every count above an edge, and the body of nature is the wide one (DERIVED from the fixed point and computed, GAMEBOARD; the identification is a decision of the model) [1, the two branches][3, the body of nature is the cloud].

The cloud, the body of nature. The wide branch is the continuum's Pekar body [38]: the count is spread over many Nodes, with a radius of about $2\Gamma/C$ Links for a count C . Its three masses agree to the order $(C/\Gamma)^2$: the well it sources, the sum over its Nodes of $D_i \text{ div } T$; the count it resists with, its inertia from the momentum reading; and the weight it falls with, from the momentum identity's force term in a content gradient. It falls as a free quantum falls. Table 5 gives the three along the branch on an open 41-cube at $\Gamma = 6,000$, with the counts in the vacuum's share at the Node, which for a cloud is the implementation's count to a percent [1, the rows against nature, (u)].

The two lighter clouds lean on the box's faces, so their true radii are larger and their numbers nearer 1 than Table 5 shows. So the universality of free fall [36] holds in the model for extended bodies to the order $(C/\Gamma)^2$, and the better the larger Γ is against the body's count. In integers a cloud is sparse: at 1,500 quanta in all there are 0.3 quanta at the densest Node, and the record shared by ten thousand Nodes carries them.

The compact pixel, collapsed matter. The narrow branch is one Node and its tail: a standing record above the top of its family's band in the wells its own form lays, with its count at one Node near the horizon and over its tail below it. Its three masses part by the binding, since a content

Table 5 The cloud, the wide branch of the binding row's fixed point, along its branch (DERIVED from the fixed point, a reading of the algebra and no click; GAMEBOARD): the total count, the rotation $2 \cos \omega_b$ against the band's top 1.3333, the rms radius in Links, the count at the centre Node, the well and the inertia over the count and over each other, and the fall against a free packet's. An open 41-cube at $\Gamma = 6,000$, matter [4000, 6000], $T = 2^{15}$, under the massless field alone; the counts in the vacuum's share at the Node, which for a cloud is the engine's count to a percent.

Total count	$2 \cos \omega_b$ (top 1.3333)	rms radius	Centre Node	Well / count	Inertia / count	Inertia / well	Fall / free
4,000	1.3509	8.7	2.6	0.985	1.061	1.077	0.92
3,000	1.3437	10.3	1.0	0.991	1.037	1.047	0.95
2,000	1.3383	11.5	0.4	0.995	1.022	1.027	0.97
1,500	1.3359	12.0	0.3	0.996	1.016	1.020	0.98

Table 6 The compact pixel, the narrow branch and collapsed matter, from the edge to near the horizon (DERIVED from the fixed point with the tension, a reading of the algebra and no click; GAMEBOARD): the count at the Node, the total count, the well it sources and the inertia, each also over the count, and the fall against a free packet's; an empty cell is a number the law's document does not give. The counts in the vacuum's share at the Node; the engine's count is about three times them along this branch.

Count at the Node	Total count	The well	Well / count	The inertia	Inertia / count	Fall / free
1,650 (the edge)	2,541	1,769	0.70	2,642	1.04	
2,000	2,640	1,637	0.62	2,621	0.99	0.27
2,500	3,005	1,636	0.54			
3,000	3,449	1,692	0.49	3,105	0.90	0.11
4,000	4,413	1,860	0.42			
4,500	4,917	1,967	0.40	4,069	0.83	0.03
5,000 (near the horizon)	5,440	2,114	0.39	4,517	0.83	

gradient reaches its quanta only through its tail, where its own well has taken most of the Link's pace. Table 6 gives them along the narrow branch, with the counts in the vacuum's share at the Node; the implementation's count is about three times these along this branch [1, the rows against nature, (u)].

A free quantum of matter at rest has a well equal to its count. A compact pixel's well is below its count everywhere in its window, 0.70 of it at the edge and 0.39 at the horizon: the binding lightens it, the more the deeper. Its inertia lies near its count and not its well, and its fall is suppressed, to none at the horizon. So a compact body sources gravity with its well, resists with its count and falls with its tail. Gravity through the pace is not a metric for a compact body: a metric would move a body and its binding together, while here the gradient moves the record and leaves its wells to follow. This is the model's statement about collapsed matter and no finding against nature, since nature's bodies are extended (Fig. 6). Between the two branches stands a saddle, where the total count falls as the binding grows, the Vakhitov and Kolokolov reading $dC/d\lambda < 0$ with $\lambda = 2 \cos \omega_b$ [41] [1, the three masses of a body; the two branches].

The electron and the proton. The electron is one quantum of the matter family laid with a sense, a message and not a body, stable because one quantum does not disperse into fractions; it needs no band of its own. The proton is a cloud of 1,836 quanta, so $m_p/m_e = 1,836$ is a ratio of counts, and it fixes no Γ . The model does not derive 1,836: it has one scale, Γ , and pure numbers of order one, so every count it derives is a fraction of Γ or a small number, and the roads tried give none. The proton's count is therefore an input, declared in the world file as every body's count is (ASSUMPTION) [1, the electron needs no band of its own; the rows against nature, (t)].

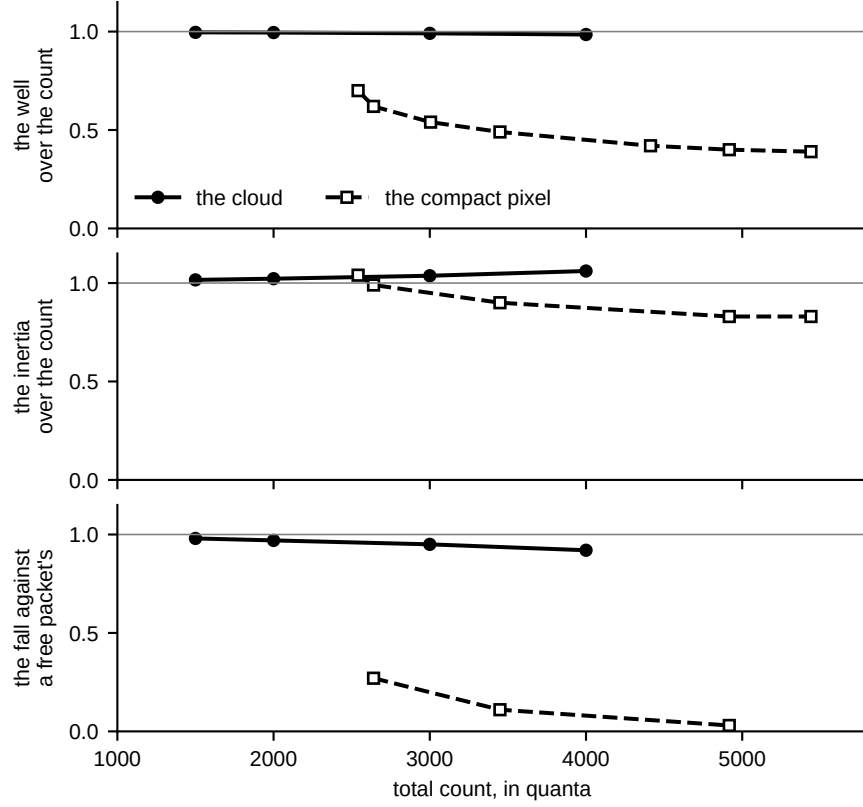


Fig. 6 The three masses of a body along its two branches, from the documents' rows (DERIVED from the fixed point, a reading of the algebra and no click, GAMEBOARD; the counts in the vacuum's share at the Node), top to bottom: the well over the count, the inertia over the count and the fall against a free packet's, for the cloud (the wide branch, filled marks) and for the compact pixel (the narrow branch, open marks), each against its total count (Tables 5 and 6). For the cloud the three go to one and the fall to a free quantum's the wider it is; for the compact pixel they part by the binding. Drawn from the rows of the law's document; no run.

The lightest body, and the binding range. Under the binding field a cloud stands only within the range R . Under $[600, 601]$, $R = 10$, the cloud of 4,000 stands and the clouds of 1,836 and 1,000 do not; the binding field holds no cloud lighter than about 1.5 times $2\Gamma/R$, so there is a lightest body. The lightest cloud is the lightest baryon, the binding field's own quanta are the mesons, and the electron is never a body. The cloud of 1,836 is not bound under $R = 10$ or $R = 15$ and is bound under $[2400, 2401]$, $R = 20$, with $2 \cos \omega_b = 1.33394$, an rms radius of 11.9 Links, its well 0.997 and its inertia 1.011 of its count and its fall 0.988 of a free packet's. So the file's pair is $[2400, 2401]$, and the range is fixed by the proton (COMPUTED, GAMEBOARD) [1, the rows against nature, (w)].

A second heavy family, and the proton's size. The fall of a free packet at rest in a content gradient g depends on its pair's gap $\epsilon = (\text{den} - \text{num})/\text{den}$, as $a/g = (1 - \epsilon)/(3(1 - \epsilon/2))$, exactly from the band: 0.2667 at $[4000, 6000]$ and $1/3$ for a light pair. So the universality of free fall holds within one family, and two families with different gaps would fall differently by up to a quarter, where nature's materials fall alike to 10^{-13} on the ground [36] and to 10^{-15} in orbit [37] (DERIVED, CLICKS). A body is never smaller than its quanta's Compton length, $0.577/\omega_0$ Links, so the model's proton, a cloud of the electron's family, is at least 0.69 Links across, where nature's proton is

460 times smaller than the electron's Compton length. A heavy family small enough to make a compact proton breaks the fall's universality; a light one gives the fall and not the size. Within Rule3, with the well as the binding, no universe has both nature's sizes and nature's fall (DERIVED, GAMEBOARD). Both sizes are lattice numbers, so the statement can be tested only by clicks, from short matter waves scattered off a cloud onto a screen [1, the rows against nature, (v) (2); what is open, items 1 and 3].

The charged cloud, and the neutron. A cloud whose every quantum carried charge would have a hill of the charge's field equal to its gravity well and would not bind; the proton stands as a cloud of 1,836 quanta with a total W of one unit, one mode whose rotation is a thousandth elliptical. The charged cloud is bound a little less than a neutral one of the same count, by its own hill, so it is heavier by about 10^{-5} of its mass; nature's neutron, the neutral one, is the heavier, by 1.4×10^{-3} . The model's difference is opposite in sign and smaller (DERIVED, a finding against nature; GAMEBOARD) [1, the two branches; the rows against nature, (z)].

7.4 A body's clicks

What a detector reads of a body follows from the body's record at its Nodes, the readings of Section 4.3 and the paces, with no further line; every formula below is a derivation with its status and its fence, none is a run's result, and the bridge to nature is a body's clicks alone (Section 1) [3, the mathematics of the bound bodies and how it bears on the clicks].

A steady rotation writes no light. A body at rest with the plane record $re = A \cos \omega_b t$ and $im = eA \sin \omega_b t$, e its ellipticity, has the Wronskian $W = -eA^2 \sin \omega_b$ at every interval, so the write of the holder of the sign is a constant source whose field stands at its rest: a static level, no travelling events, no click anywhere; the body's charge is $eA^2 \sin \omega_b / T$ quanta per interval of source, $1 / \sin \omega_b$ of its count for a circular quantum. A body in uniform motion radiates nothing either: its retarded level follows it and carries no travelling events. A breathing body, $A(t) = A_0(1 + \delta \cos \Omega t)$, has the Wronskian $W(t) = -eA_0^2 \sin \omega_b$ times $(1 + 2\delta \cos \Omega t + \delta^2 \cos^2 \Omega t)$, so the light born carries the breathing's frequency Ω , and 2Ω at the second order, and not the rotation ω_b , with $2\delta eA_0^2 \sin \omega_b / T$ quanta per interval in its first harmonic; a far detector's click rate is that current's share through its front, the density times the group velocity, and a cloud's Ω is the difference of its bound modes' rotations. So a symmetric body at rest clicks nowhere, and a body is seen by what it absorbs and by what it writes when it breathes or moves (DERIVED; the Wronskian and the source GAMEBOARD, the far detector's rate CLICKS). Beside it: a steady current radiates nothing and an oscillating one radiates at its own frequency [1, light is born by the write].

The invariant, and the count's drift. For one Node of a body, $a_{\text{next}} + a_{\text{before}} = 2 \cos \omega(t) a_{\text{now}}$ with ω changing slowly has the solution $a_t = C \cos(\sum \omega + \phi) / \sqrt{\sin \omega(t)}$, so $A^2 \sin \omega_b$ is invariant, the discrete adiabatic invariant of Rule3's line, while the form $D = A^2 \sin^2 \omega_b$ and the share are not; the recurrence stepped over 20,000 intervals with ω lowered from 0.841 to 0.600 keeps $D / \sin \omega$ to 4×10^{-5} while D falls by a quarter. The share in quanta at a Node, $(3\Gamma^2 / \sum_a p_a^2) A^2 \sin^2 \omega_b / T$ (Section 4.4), therefore moves under a moving well as

$$\frac{N(t_2)}{N(t_1)} = \frac{\sin \omega_b(t_2)}{\sin \omega_b(t_1)} \cdot \frac{\sum_a p_a^2(t_1)}{\sum_a p_a^2(t_2)}, \quad (13)$$

with no current through any Link: the first factor is the rotation's own change under the clock's pace, a deepening well lowering ω_b and the count read with it, the second the Link paces' weights in the share. For a mode taken from the band's top to the compact pixel's rotation the first factor

is 0.94 and the second, at the pixel's centre, about 5, so the share at a Node reads several times its vacuum share (THEOREM for the invariant, DERIVED for the drift, COMPUTED for the numbers; GAMEBOARD). The invariant is what nature calls the number of quanta, E/ω , and the share is nature's energy; the two part by $\sin\omega_b$ and the paces' weights, and the model's count is the share by declaration (Section 4.4). The same identity is Planck's relation on the lattice: a standing record carries the share $A^2 \sin^2 \omega/T$ per Node and the invariant $I = A^2 \sin \omega/T$, so share = $I \sin \omega$ at every Node and wave number, $E = \hbar\omega$ with T for $\hbar$ and $\sin \omega$ in place of ω . The credit counts the share: a detector of the model credits two beams of equal power the same number of quanta, where nature's photon counter credits the longer wavelength more, by ω_1/ω_2 , 1.9 between the wavelengths 8 and 16 Links. The two readings agree at one wavelength and part between wavelengths; the model's unit is a declaration, the one line that would make the number the unit is the invariant, which is not in the law, and a world of two wavelengths with bodies as counters would read the difference (DERIVED; the unit ASSUMPTION; CLICKS) [1, the count is the record's share; the paces' own change moves the share with no current].

The clock's click rate. A bound body's rotation reads the content once: at rest $\cos\omega_0(U) = \cos\omega_0 + 2gU$ to first order, with $g = (\text{den} - \text{num})/\text{den}$, so its rate falls by $f(\omega_0)U$ with $f(\omega_0) = 2(1 - \cos\omega_0)/(\omega_0 \sin\omega_0)$, 1.063 at [2, 3] and 1 as the gap closes; the rate $1 - U$ of a clock that follows the clock's pace alone, p_0/Γ , is the small-gap limit, and a clock that reads the content once is a bound body's rotation and no light clock. A light clock between two faces falls by $2U$, the Link's pace read twice (Section 10.3); a moving body's tick is Lorentz's factor at its own light speed (Section 8, (e)) (DERIVED, CLICKS).

The resonance. A bound record at the rotation ω_b , $2 \cos \omega_b$ above its band's top, reads a light level $b_L \cos \omega_L t$ into its content at the weight 1, the content x_0 at its Node otherwise static and the body not yet writing back. At $k = 0$ the Node's line is $a_{\text{next}} + a_{\text{before}} = 2 \cos \omega(t) a_{\text{now}}$ with $2 \cos \omega = 2 - 2g(1 - 2x + 2x^2)$, so the light modulates the rotation by $\epsilon = 4g(1 - 2x_0) b_L/\Gamma$. Writing $a_t = \text{Re}[z_t e^{i\omega_b t}]$ with z slowly varying, the resonant part of the modulation at $\omega_L = 2\omega_b$ gives $z' = -i\epsilon e^{i\vartheta} \bar{z}/(4 \sin \omega_b)$, whose eigenvalues are real: the amplitude grows at $\mu = \epsilon/(4 \sin \omega_b)$ per interval, the form and the share at 2μ ; detuned by $\Delta = \omega_L - 2\omega_b$ the exponent is $\sqrt{\mu^2 - \Delta^2/4}$, so the body takes light only inside the full width $\epsilon/\sin \omega_b$, and at $\omega_L = \omega_b$ only at the second order in ϵ . This is the one-mode case: the same modulation multiplies the record and mixes its frequencies, so a body with two bound modes $\omega_i < \omega_j$ takes light at $\omega_L = \omega_j - \omega_i$ and at the sum, the line between two bound states, of which $2\omega_b$ is the degenerate case $i = j$. The recurrence stepped at the compact pixel's rotation, $2 \cos \omega_b = 1.4317$, grows at 0.00461 per interval at $\epsilon = 0.013$ against the formula's 0.00465, with the detuned exponents as the formula gives them; a level $b_L = 60$ at $\Gamma = 6,000$ gives $\epsilon = 0.013$ in the vacuum and 0.006 at the pixel's centre. The meeting's question for a bound body, whether the past and the future meet at it, is whether the arriving record's frequency lies in this window; outside it the light passes and the body's share does not rise. The transit's sinc^2 of width $2\pi/T_{\text{ov}}$, with T_{ov} the overlap's duration, multiplies the line, so its width is the larger of the two. The read's side is derived; the fraction absorbed, with the body's write back, is computed when the fixed point with the write is, and the two-mode transfer rate is to be derived (DERIVED for the one-mode resonance, its growth and its width, COMPUTED for the numbers; the record under the light GAMEBOARD, the body's share rising at its front as a declared instrument CLICKS). Beside it: the parametric resonance of an oscillator modulated at twice its frequency, with the growth $\hbar\omega_0/4$ for the depth h , and Rabi's power-broadened line, whose width is the coupling itself [1, the rows against nature, (y)].

A body's count in clicks. A body's count is its share summed over its Nodes, what a full absorber credits, and under a moving well it drifts by Eq. (13) (DERIVED, CLICKS). The formulas of clicks stand together in Table 2 with their statuses.

7.5 What a stone is

From one quantum to a stone, in the model's terms (DERIVED where it follows from the lines and ASSUMPTION where a decision is named; no body has yet been run, Section 9). (i) *A quantum of matter* is a record of the matter pair $[m, \Gamma]$ at rest, rotating at $\cos \omega_0 = m/\Gamma$; free, it disperses by its band and carries one quantum in its share. (ii) *A body* is many quanta of one family whose record stands: its form is written each interval into the binding field at its own Nodes, the field rests at the screened sum with the range R , every Node of the record reads that level into its paces, and in those paces the record's rotation lies above the top of its family's band, so no part of it can travel away. A body is the fixed point of that loop, the record whose own well returns it. (iii) *A nucleus* is bodies within the binding field's range of one another, $R = 20$ Links, bound by the screened sum, the model's strong force, with the field's own travelling events as its mesons. *An atom* is a charged cloud with an electron held in the well of the charge's field it writes; the two signs of the charge wait for Hypothesis 8. (iv) *A stone* is many such bodies standing in one another's wells, within reach of each other's tails: who is bound to whom is read from the nesting of the levels at the Nodes and never declared, and the whole is one body for the clicks, since one click at one place tells that the whole acts as one. A stone moves by Rule3 alone, its quanta following its record's current with no record of its position. Its clocks are slowed alike by the well it stands in, and it falls as a cloud falls, as a free quantum falls to the order $(C/\Gamma)^2$. At rest and symmetric it clicks nowhere. It is seen because the bodies at its surface act as instruments in the meeting's form, absorbing light at their rotations' resonance and writing light of their own when they breathe or move (Section 5.4). It sources gravity with its well and resists with its quanta, and its three masses are one to the order $(C/\Gamma)^2$ because its bodies are clouds. What the model has not shown of a stone: no body has run under the lines of this paper, and the proton's size stands as a finding against nature [1, one thing, two faces; the click joins and parts][3, what a body is].

7.6 Clusters: how they enter, and what each thing represents

How a cluster enters. The model writes no line for a cluster. One assumption names the universe the model describes: every body is bound, no body is free, and a record that is in no body is a message between two clicks, the one that gave it and the one that will take it (ASSUMPTION). Under it a body is bound by the three fields its family reads and by nothing else, and a cluster is bodies standing in one another's wells, bound by the same three fields and clicking at one another. Between two bodies there is nothing but the click, the light of the holder of the sign and the two real fields: no record leaves its body, since its tail falls off by $e^{-\kappa}$ per Link; a quantum crosses a Link whole at a click and nothing else does; the overlap of two tails within reach is read by the currents; and light and the fields carry the rest. A Node is a body's Node or vacuum, it holds its records' levels and remainders and nothing else, and its count is a reading of them (DERIVED, GAMEBOARD) [1, the bound body is one Node; the universe is bound; the algebra of clusters].

What each thing represents in a cluster. (i) *The three levels at a Node are the nesting.* The binding field's level at a Node is the matter's tail, the level of the holder of the sign is the charge's well and the massless field's level is gravity's well: they are the three kinds of binding by rank, and the size of a level is the depth in its kind. A reading of the three levels at a Node therefore names the clusters the Node belongs to, a diagnostic and no click (DERIVED, GAMEBOARD). (ii) *Three*

kinds of binding, not three levels. The three ranks give three kinds of binding: quanta into a body by the binding field's screened well, with the range R ; bodies into a charged cluster, an atom or a molecule, by the field of the holder of the sign; charged clusters into a gravitational cluster by gravity's $1/r$ well. The third kind nests without bound, so the count of gravitational levels is the universe's history and no number of the model. (iii) *The hierarchy is recursive.* A cluster of clusters is the same structure one level up: its members are bodies clicking at one another as a body's Nodes do, and a level $n + 1$ is bound by the same inequality as the level 1, with the cluster's total count over its members' spacing in place of a body's count over one Link. No number bounds the depth but the horizon at a Node, the content read there reaching $\Gamma/2$, at which a cluster's centre is a frozen Node (Section 3.4). (iv) *The click is the pointer.* The click is the only thing that passes between a cluster and its neighbour and between a level and the next. Inside a cluster everything is seen, a reading of the lattice being the local language of its own record; beyond its tail only a click is seen, a detector's report. So the observer is a cluster: a Node reads six Ports, a cluster's record is what its Nodes carry, and what it knows of the world beyond its tail is its clicks, which is the model's form of the statement that an observer sees only clicks (Section 1) (DERIVED, CLICKS). (v) *The screen is a cluster.* In the rule's own universe everything that stands on the lattice is a body or a cluster of bodies, the source, the screen, a slit's wall, a mirror and a detector alike; a face of the GameBoard is no body, and a click on a face is no click. The implementation's two gates declare their walls, faces and detectors as instruments instead (Section 9), each a body's role written without its record, the stand-in of a body until the bodies run (ASSUMPTION). (vi) *The click joins and parts.* Two clusters whose tails overlap read each other by the currents. The cross current of two standing records at the rotations ω_1 and ω_2 is a sum of cosines of $(\omega_1 - \omega_2)t$ and $(\omega_1 + \omega_2)t$ with no constant term, so its mean over the beat is 0 at every Link: at fixed paces no quantum passes at the first order, the counts swing at the beat about their values, and a net transfer is of the second order, the well moving with the count, or radiative, by the messages. The click's reach is where the Link's current $a_1 a_2 e^{-\kappa d}$ falls under T , at $d = \ln(a_1 a_2 / T) / \kappa$ Links; beyond it there is no click, and the two are parted. Where the smaller gives to the larger until its count falls under the lightest body's (Section 7.3), it dissolves into messages the larger takes. A join and a parting are runs of local clicks, after which the smaller's levels go on as messages; no record opens or closes anywhere, and there is no non-local act (DERIVED; the current GAMEBOARD, the reach CLICKS). (vii) *The two readings of Γ are two levels.* Every level of the hierarchy has its own clock unit and its own interval, a level's interval being the period of the record one level below it. So the proton to electron ratio is a count's ratio within a cluster and reads no Γ , while a universe whose Γ steps (Hypothesis 2) reads its age in the top level's Γ ; the tick ratio between two levels is the cluster's period over its member's, $P_2/P_1 = e^{\kappa d}$ on the narrow branch, 7.6 for two compact bodies of 2,000 quanta two Links apart at $\Gamma = 6,000$ (DERIVED, COMPUTED for the number; GAMEBOARD). (viii) *The couplings are at the edge, and the one integer.* Every coupling is the edge the rule allows and no number: every divisor is 1 but gravity's E_g , and the universe of clusters carries Γ and T and nothing else. Collapsed matter has one stable count, the frozen Node, the endpoint of accretion on the narrow branch; the body of nature is the cloud, and a stone is clusters of clouds (Section 7.5) (ASSUMPTION for the divisors, DERIVED for the rest; GAMEBOARD) [1, the levels at a Node are the nesting; three kinds of binding, not three levels; the hierarchy is recursive; the click is the pointer; the observer is a cluster; the screen is a cluster; the click joins and parts; the two readings of Gamma are two levels; the couplings are at the edge; the stable pixel is one][3, what a body is; the rule's own universe is the universe].

8 The forms of nature the rule arrives at

Each form below follows from the rule's coefficients, read by the geometric optics of a packet's band, before any run; the form of nature beside it is on the comparison side and is never an input. The coefficients contain no metric, no Lorentz transformation, no potential and no force. They contain the paces of Section 3.4, and two features of the paces were chosen with a known result: the factor two on the Link, fixed by light's bending, and the clock's second order, adopted with the perihelion known. Where a form rests on either, it is marked FITTED. $U = c/\Gamma$ denotes the model's potential at the point named, the clock's share per level, and U_b its value at the impact parameter b [1, the rows against nature].

Two potentials. The forms are stated in two potentials, and the paper carries both. The model's own potential is the clock's share, $U = c/\Gamma$, the content x that the paces read; no instrument reads it. Every test of gravity takes GM from Kepler's law of the orbits of matter and c from light, so it reads the potential $U_K = GM_K/(c^2 r)$. In the model a free packet of matter of the pair [num, den] falls in a content gradient at $a = [(1 - \epsilon)/(3(1 - \epsilon/2))] |\nabla x|$ with $\epsilon = (\text{den} - \text{num})/\text{den}$, exactly from its band (Section 7.3); with $\cos \omega_0 = 1 - \epsilon$ and $c^2 = 1/3$ this is $a = c^2 [2 \cos \omega_0 / (1 + \cos \omega_0)] |\nabla x|$, so the orbits of matter measure

$$U_K = \frac{2 \cos \omega_0}{1 + \cos \omega_0} U, \quad 0.800 \text{ at } [2, 3], \quad 1 \text{ as the gap closes}, \quad (14)$$

and an orbit computed from the band's rays gives 0.797 at [2, 3], the finite-potential remainder, and 0.9999 at [5999, 6000]. Each form below is written first in U , where the two fitted inputs give Einstein's coefficients, $\gamma = \beta = 1$ in the post-Newtonian words, and then in U_K , where the same algebra gives $\gamma_K = \text{den}/\text{num}$, $\beta_K = (\text{den} + \text{num})/(2\text{num})$ and $1 + \alpha_K = \tan \omega_0/\omega_0$: 1.5, 1.25 and 1.329 at [2, 3], 1.0002, 1.0001 and 1.0001 at [5999, 6000], and 1 as the gap closes. The two columns are one algebra, and nature's tests read the second (DERIVED, CLICKS) [1, the rows against nature][3, general relativity in the law].

- (a) *The redshift* (DERIVED, CLICKS). Two clocks of one family at the levels c_1 and c_2 tick in the ratio of the root of $(1 - 2U_1 + 2U_1^2)/(1 - 2U_2 + 2U_2^2)$, which is $1 - (U_1 - U_2)$ to first order, for a band whose rotation follows the clock's pace alone. The band's own slope adds a factor: at the content x a record of the pair [num, den] at rest rotates at $2 \cos \omega = 2 - 2((\text{den} - \text{num})/\text{den})((1 - x)^2 + x^2)$, so a bound rotation shifts by $f(\omega_0)x$ with $f(\omega_0) = 2(1 - \cos \omega_0)/(\omega_0 \sin \omega_0)$, 1.063 at [2, 3] and 1 as the gap closes: the gravitational redshift is $z = f(\omega_0)U$, and the rate $1 - U$ is its small-gap limit. Against Kepler's potential it is $z = (1 + \alpha_K)U_K$ with $1 + \alpha_K = \tan \omega_0/\omega_0$, 1.329 at [2, 3], where nature's clocks give $|\alpha| < 10^{-4}$ [36]. Light written by a breathing body in the well leaves with the body's rotation and keeps it in flight, since a static content changes no frequency, so a far detector reads the emitter's slow clock. Beside it: Einstein's gravitational redshift [36].
- (b) *The bending of light* (COMPUTED, FITTED; CLICKS). At a Node of content x light's index is $n = 1/(1 - 2x)$, so $n - 1 = 2x$, because the Link's pace reads the content twice. Far from a body its field is the massless field's rest, $L(r) = 3\sigma_{\text{tot}}/(4\pi r)$, with σ_{tot} the body's wells per interval, so $n - 1 = 3\sigma_{\text{tot}}/(2\pi \Gamma r)$, and a beam passing the body at the closest distance b turns by the transverse gradient of the index integrated along the path:

$$\alpha = \int_{-\infty}^{\infty} \frac{\partial n}{\partial b} dl = \frac{3\sigma_{\text{tot}}}{2\pi \Gamma} \int_{-\infty}^{\infty} \frac{b dl}{(b^2 + l^2)^{3/2}} = \frac{3\sigma_{\text{tot}}}{\pi \Gamma b} = 4U_b, \quad U_b = \frac{3\sigma_{\text{tot}}}{4\pi \Gamma b}, \quad (15)$$

twice the angle $2U_b$ that a Newtonian pull of the same potential gives. Computed by the geometric optics of the band from the coefficients themselves, the deflection is 2.02 times

Newton's $2U_1/b$, with the clock to first or to second order; with the Link's pace lowered once instead of twice it would be 1.01 times Newton's. The factor two is therefore the fitted choice of Section 3.4, and this item checks consistency rather than predicting. What the model adds is the comparison with a matter clock: the bending is $2x$ and the matter clock's redshift is $1.063x$, so light is bent by 1.9 times a matter clock's share, where general relativity gives 2; the 1.063 is the band's slope at the matter pair and was not chosen. Against Kepler's potential the bending is $(1 + \gamma_K) 2GM_K/(c^2b)$ with $\gamma_K = \text{den}/\text{num}$, 1.5 at $[2, 3]$, where Cassini's tracking gives $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ [36]. Beside it: Einstein's $4GM/(c^2b)$ [36].

- (c) *The delay of light* (DERIVED, FITTED; CLICKS). Light's coordinate speed at the content U is $(1 - 2U)/\sqrt{3}$, so its index is $1 + 2U$, and a signal grazing a body at the impact parameter b between the distances x_1 and x_2 , with $U = \mu/r$, is late by $2\sqrt{3}\mu \ln(4x_1x_2/b^2)$ intervals: Shapiro's delay with $\gamma = 1$, the same integral whose transverse derivative is the bending, from the same factor two. Against Kepler's potential it carries the same γ_K . Beside it: Shapiro's delay [46].
- (d) *The perihelion* (COMPUTED, FITTED; CLICKS). An orbiting emitter at the semi-axis a and the eccentricity e advances by $6\pi U_p$ per orbit, with U_p the potential at $a(1 - e^2)$. By the same geometric optics the matter pair's periapsis advances by $7\pi U_1/(a(1 - e^2))$, 1.17 times Einstein's value, with the clock to first order, and by $6\pi U_1/(a(1 - e^2))$, Einstein's value, with the clock's second order; the bending stays $4U_b$ under both. The second order was adopted with this result known. Against Kepler's potential the advance is $(2 - \beta_K + 2\gamma_K)/3$ of Einstein's with $\beta_K = (\text{den} + \text{num})/(2\text{num})$, 1.25 at $[2, 3]$, so 1.25 of Einstein's there (1.26 on a computed orbit), where the planets' and the Moon's orbits give $|\beta - 1| < 10^{-4}$ [36]. Beside it: Einstein's $6\pi GM/(c^2a(1 - e^2))$ [36].
- (e) *The moving clock: Lorentz's factor* (DERIVED, CLICKS). A body moving at the velocity v has its tick lengthened by the dispersion of its own bound band. The tick's factor is $f = \Omega(k)/\omega_0$, with $\Omega(k) = \omega(k) - k\omega'(k)$ the rotation at the moving centre, the phase rate less k times the group velocity, and $\omega(k) = \omega_0 + \Delta(1 - \cos k)$ the body's bound band, with ω_0 its rest rotation and Δ its packet's top velocity, both fixed by the file's counts and pair. To second order $f = \sqrt{1 - v^2/c_b^2}$ with $c_b^2 = \Delta\omega_0$: Lorentz's factor at the body's own light speed. It is read as the mean of the front and back click periods $P_v(1 \mp v/u)$ at two resting detectors, with u light's group velocity. No boost is among the 48 symmetries, so Lorentz's factor appears as a property of the clicks and not as a symmetry of the lattice. Nothing here was fitted. Beside it: Lorentz's factor, at the body's own light speed rather than the lattice's [3, the click].
- (f) *The force between parallel moving charges* (DERIVED as a finding against nature under the rule as it stands; HYPOTHESIS for the form that gives nature's; CLICKS). Under the plain read, the rule as it stands, the level of the holder of the sign enters a charged reader's paces as every level does, and a reader moving at v in a static level has its rotation at the wave number k fall with the level faster than a resting one's, because the Link's hopping weakens where the level rises: the force on it, $-(d\omega/dx)(dx/dr)$, grows by the factor $(\sin \omega_0/\sin \omega(k)) [1 + (2\text{num}/(3\text{den}g))(1 - \cos k)] = 1 + 3(2 + \cos \omega_0) v^2/(2\cos \omega_0) + O(v^4)$, with $g = (\text{den} - \text{num})/\text{den}$; this is $1 + \frac{3}{2}\beta^2$ for a small gap, with $\beta = v/c$, and $1 + 2\beta^2$ at $[2, 3]$, for a hollow and a hill alike. The moving source's level is light's own record at c , whose retarded solution, Liénard and Wiechert's scalar, has a transverse gradient γ times the static one. So two like charges moving side by side repel, for a small gap, with

$$F = \gamma \left(1 + \frac{3}{2}\beta^2\right) F_C = (1 + 2\beta^2) F_C + O(\beta^4) \quad \text{under the plain read,} \quad (16)$$

$$F = \gamma(1 - \lambda v^2) F_C \quad \text{under the rotation of Hypothesis 8,}$$

where nature's two parallel like charges repel with $F_C/\gamma = (1 - \frac{1}{2}\beta^2) F_C$: the correction has

the wrong sign, and magnetism is not derived from the rule as it stands. Nature's force on a charge moving in a static field is qE , independent of v , which the plain read does not give either. What gives nature's form is Hypothesis 8: the holder of the sign acts on a plane record by a rotation, its time part turning the record by the angle θ_0 per interval and each of its three Link parts turning the arrivals through the $\pm a$ Ports by $\mp\theta_a$, Peierls' phase on the Link. A plane record then obeys the vacuum's band with $\omega \rightarrow \omega - q\theta_0$ and $k_a \rightarrow k_a - q\theta_a$, q the record's own sense, and with the kinetic wave number $\mathbf{K} = \mathbf{k} - q\boldsymbol{\theta}$ and $\mathbf{v} = \partial\omega/\partial\mathbf{K}$ the rays of every band obey, as an identity of geometric optics, $d\mathbf{K}/dt = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$ with $\mathbf{E} = -\nabla\theta_0 - \partial\boldsymbol{\theta}/\partial t$ and $\mathbf{B} = \nabla \times \boldsymbol{\theta}$: the Lorentz force as the lattice's own identity, the time part the scalar potential, the three Link parts the vector potential, and $-\partial\boldsymbol{\theta}/\partial t$ induction. The time part is sourced by the Wronskian W and the Link part a by the Wronskian's current through the a Ports, which for a circular record moving at v_g is $3 \text{ den } v_g |W|$, the density times the group velocity; both step on the pair $[1, 1]$, so a source in uniform motion lays the retarded levels, the time level γ times the static one across the motion and the Link level λv times the time level along it, with λ the Link part's write per unit of current over the time part's per unit of density. For two like charges side by side the sum is $F = \gamma(1 - \lambda v^2) F_C$, and nature's F_C/γ needs $\lambda = 1/c^2 = 3$. The Link part's wall decides λ : written through $E_s \text{ den } T$, one axis's measure of a current, the Link level is $3v_g$ times the time level and $\lambda = W_c/(\text{den } T) = 3 = 1/c^2$ exactly, the quantum's measure of a current over one axis's measure being the inverse square of light's speed, with no inserted number; written through the tension's wall $E_s W_c$ it is $\lambda = 1$, and the magnetic part is a third of nature's. The wall is a number the rule does not fix, and nature's own measurement of it is Ampère's force against Coulomb's, already $1/c^2$ (EXPERIMENT). At $v = 0.1c$ and $0.5c$ the ratio F/F_C is 0.995 and 0.866 under the rotation at $\lambda = 3$, 1.002 and 1.058 at $\lambda = 1$, and 1.020 and 1.588 under the plain read, so one computation of two charged bodies moving in parallel tells the three apart. Derived under the hypothesis: the parallel case, the Lorentz identity and induction; nothing of it is in the rule as it stands. Beside it: the Lorentz force between parallel currents, and Biot and Savart's force read as the electric force in the charges' own frame $[1, \text{ the rows against nature, (b2)}][3, \text{ the current's read; magnetism is in the world of clicks}]$.

- (g) *The two signs of the charge* (open; CLICKS). Under the plain read, which is the law today, the level of the charge's field enters every charged reader's paces in the same way, so the model gives the size of the electric pull but not its dependence on the reader's sign: like and unlike charges are told apart only by the rotation of Hypothesis 8, which is not yet in the law. The model has no line in Q/M , and the response to a gradient is the mode's own, so an electron and a proton read one field alike. Beside it: Coulomb's force, with like charges repelling.
- (h) *De Broglie's relation* (DERIVED; GAMEBOARD for the band, CLICKS for the fringe spacing and the time of flight that read it). A free record of a matter family travels only where its band holds its rotation, $\text{num}/(3 \text{ den}) \leq \cos\omega_b \leq \text{num}/\text{den}$. Its wave number is the band's, $\cos\omega_b = (\text{num}/(3 \text{ den}))(2 + \cos k)$, its group velocity is $v = (\text{num}/(3 \text{ den})) \sin k / \sin\omega_b$, and to second order $k = \omega_b v / c_m^2$ with $c_m^2 = \omega_0/(3 \tan\omega_0)$, the family's own light speed. Beside it: $\lambda = h/p$ with the rest rotation as the mass.
- (i) *The fall* (DERIVED, CLICKS). In a pace gradient a free packet or a cloud accelerates at $a = \Delta |d\omega_0/dc| (c_+ - c_-)/D$ toward the well, with $c_{\pm}$ the read level at its two faces D Links apart. So bodies with the same $\Delta d\omega_0/dc$ fall alike, and a cloud falls as a free quantum falls to the order $(C/\Gamma)^2$ (Section 7.3), while a compact pixel falls by its tail alone, more slowly. The universality holds within one family, and the model has one family of matter. Beside it: Galileo's fall and the equivalence principle [36], which hold in the model for extended bodies and not for collapsed matter.
- (j) *Newton's pull and orbits* (DERIVED, CLICKS). The massless field rests at Poisson's solution with

Table 7 The numbers of one body, the 2,000 pixel at $\Gamma = 6,000$ (COMPUTED from the law's lines, no run; CLICKS): at each distance from the pixel, the field in levels, the clock's share z and the bending of light at that impact parameter, under the binding field [2400, 2401] and under Poisson's rest; an empty cell is a number the law's document does not give.

Distance from the pixel	The field, in levels		The clock's share z		The bending at that impact parameter	
	[2400, 2401]	Poisson	[2400, 2401]	Poisson	[2400, 2401]	Poisson
1 Link	373	386				
2 Links	189	205				
3 Links	117	133			4.9 deg	5.0 deg
5 Links	62	79	1.09 %	1.4 %	2.8 deg	3.0 deg
10 Links	24	40			1.2 deg	1.5 deg

the sources and at Laplace's outside a body, $c(r) = 3s/(4\pi r)$, so the acceleration falls as $1/r^2$ and a circular orbit's v^2 as $1/r$; within the range R a reader reads the binding field's screened level instead. Beside it: Newton's pull and Kepler's fall-off; a flat rotation curve would point to a missing law.

- (k) *The medium* (DERIVED, GAMEBOARD). Inside a body of pace p , light of wave number k is slowed to $v_{\text{in}} = (p/\Gamma)^2 \sin k_{\text{in}}/(3 \sin \omega)$ with $1 - \cos k_{\text{in}} = (\Gamma/p)^2(1 - \cos k)$, and is reflected whole where $p < \Gamma \sin(k/2)$; through a window moving at w the wave number is matched at the moving face, $\omega - kw = \omega_{\text{in}} - k_{\text{in}}w$. Beside it: the refractive index and the mirror. Fresnel's drag is nature's number and is not derived here.
- (l) *The round trip and Sagnac's effect* (DERIVED, CLICKS). A radiating body receding at v from a mirror at rest reads its returned clicks every $P(u + v)/(u - v)$ intervals for a write every P , with u the group velocity at the wave number that solves $\Omega(k) + kv = \omega_b$. A radiating body moving at v along a ring of N Links reads its two arms $2Nv/(u^2 - v^2)$ apart, $4A\omega/u^2$ for the ring's area A and angular speed ω to first order. Beside them: the two-way Doppler ratio and Sagnac's $4A\omega/c^2$, with the group velocity u in place of c .
- (m) *A free record keeps its wave number* (THEOREM, GAMEBOARD). Rule3 is invariant under translation, so a free record keeps its wave number and rotation exactly, and a radiating body's period is read unchanged at every distance. Beside it: Hubble's drift, which the model does not give from a fixed Γ ; its own line for an expansion is Hypothesis 2.

The numbers of one body. For a compact pixel of 2,000 quanta at $\Gamma = 6,000$ (COMPUTED, CLICKS): under Poisson's rest the bending is $0.26/b$ radians, 3.0 degrees at $b = 5$ Links and 1.5 at 10; a packet of wavelength 8 Links passing at $b = 6$ turns by 0.043 radians, a transverse shift of 1.3 Links over 30 Links of path; and a matter clock five Links from the pixel, at $x = 0.013$, runs 1.4 percent slow against a far one. With the binding field, a reader within R reads the binding field's screened level instead of the massless field's; Table 7 sets the two side by side [1, the rows against nature, (a) and (b)].

The bending stays 1.9 times the matter clock's share in both columns. Computing this ratio to three digits on the lattice, for a packet passing a pixel of 2,000 at five Links against the clock's share at the same Node, is the sharpest test the model offers: 2.00 within the rounding would confirm general relativity's ratio, and 1.9 would be a lattice correction [1, the rows against nature, (b) and (v)].

What goes in and what comes out. The model puts in one rule, the paces' read and the files' numbers. In the model's own potential three of the forms above are consistency checks of choices made with them known: the bending (b), the delay (c) and the perihelion (d). Against the potential that nature's tests read, the same three give γ_K , β_K and α_K that leave 1 by 25 to 33 percent at the

rule's own matter pair and reach 1 only as the matter pair's gap closes: nature's bounds require $\text{den/num} - 1 < 2 \times 10^{-5}$, so the matter pair is a number for experiment to fix, and the rule's own [2, 3] stands against nature here, the finding of scale seen from gravity's side (DERIVED, CLICKS). Three forms follow without any fitted input: the redshift's band factor (a), Lorentz's factor at a body's own light speed (e) and the universality of free fall for extended bodies (i). The force between moving charges (f) is not derived under the rule as it stands, and the sign of the electric force (g) is not yet derived. Where the model's number differs from nature's form at the declared numbers, the difference is the model's own and is stated: the bending's 1.9 against a matter clock, Sagnac's u against c , and Fresnel's drag, which is not derived.

9 The implementation's two gates: the two slits and Bell, derived and not run

The paper carries no experiment. Until a run has worked and been checked independently, the paper is about the algebra alone: what was derived, what was computed, and what was not. So no number from a run stands here: no count through a screen, no S of any run and no arrival interval read. What stands is two worlds of the implementation, each as the model's own derivation of what the world must give, marked DERIVED or COMPUTED and never a result. They are the implementation's two gates: a gate is a world whose every number is fixed in advance by the algebra, so that a run can only agree with it to the rounding or reveal a defect of the implementation. The two slits is the world whose number is Rule3's own propagation on the lattice, its fringe row fixed by the rule's line in real arithmetic and its arrival interval against a continuum wave's the lattice's own number. Bell's world is a gate and not an experiment: its number is fixed by the declared credit and the parts' determinism. The two slits' run enters a later version of the paper only when it has worked and been confirmed, as the proof that the integer implementation computes the law's line and as the figure of the fringe from clicks. Each expectation is written before its run and never changed after it, and nothing in a world is built to bring the right result [3, the paper; one experiment and one gate; nothing in an experiment brings the right result by construction].

9.1 The two slits: the world

The emitter is a declared message, a packet of light, the real record of the holder of the sign, and no body. Its wavelength is 8 Links along $+x$, $k = \pi/4$, with $\cos \omega = 0.9024$, a period of 14.1 intervals and a group velocity of 0.547 Links per interval, $(1 - 0.083 k^2)/\sqrt{3}$. Its levels are laid as $\text{now}_i = b e_i \cos(kx_i)$ and $\text{before}_i = b e_i \cos(kx_i + \omega)$, the record one interval earlier, with $b = 1,328$, about ten quanta per Node, under a raised cosine along x over $x = 4$ to 16, flat across $y = 4$ to 44 with edges of two Nodes. The wall is the GameBoard's own face rule declared inside it: the inner face at $x = 20$ with two gaps of three Nodes, $y = 17$ to 19 and 29 to 31, whose centres are $d = 12$ Links apart. The GameBoard is flat, $45 \times 48 \times 1$ with the third axis folded, its y faces open and its far x face receding, so that the screen, the last column $x = 44$, $L = 24$ Links from the wall, is backed by the receding face and is opaque. The screen is twelve declared detectors, each four rows of that column, the smallest region the model allows at this wavelength; the seed of the draw, 24, is declared in the expectation file. The run is 130 intervals, with the screen read over the intervals 45 to 120. The universe file is the light-only one, with the three holders of the content at the divisor 10^6 so that light's own well rounds to nothing. For a record uniform along the folded axis, light's band on the flat GameBoard, $2 \cos \omega = (2/3)(\cos k_x + \cos k_y + 1)$, is the three-dimensional band at $k_z = 0$, so the folded GameBoard is the experiment exactly (DERIVED, GAMEBOARD). In the

implementation the world is written by a builder from one design file that holds every number above, with its mode file and its expectation file beside it, under `examples/events/two_slits/` [1, the rows against nature, (g)][3, the two slits]. For the build the world gains a receding face on the source side too, so that the wall's return through the source's face never re-enters and the reading is free of the window; on that world the law's line gives 278.0 quanta through the screen, the peak of the screen's inflow at the interval 68.0 against a continuum's 61.8 and the centroid at 72.8 against 68.4, and the wings 16.4 and 12.4 against 23.7 and 20.4 (COMPUTED, CLICKS) [3, one experiment and one gate].

9.2 The two slits: what the model says it must give

Three readings of the one world, each written before any run. (a) *The fringe*. The far field's spacing is $\lambda L/d = 16$ Links. At $L = 2d$ the fringe is the near field's: the gaps' path difference reaches half a wavelength 8.7 Links from the centre and a whole one 22 Links out, so the minima fall at the rows 15 and 33 (DERIVED, CLICKS). Light's line at the vacuum's paces is $a_{\text{next}} = (\text{the six arrivals})/3 - a_{\text{before}}$, and stepped in real arithmetic on the world it gives 280.8 quanta through the screen over the window, per region from the row 0: 20.5, 19.7, 27.1, 16.2, 9.4, 41.1, 47.7, 14.6, 12.2, 27.1, 21.6 and 23.4 (COMPUTED, CLICKS). The Huygens sum of the two gaps with the obliquity factor gives 273, three percent less, with the same central fringe; the regions 3 and 8, at 16.2 and 12.2 against the continuum's 22.6 and 19.2, are the lattice's own. The implementation's integers must agree with the real line to the rounding, about a third of a quantum per region; a larger difference is a defect of the implementation and never a finding. (b) *The clicks*. The screen's shares rounded to whole quanta by the largest remainders are the expectation, and the row drawn by the shares at the declared seed is the clicks, with the draw's scatter $\sqrt{Np(1-p)}$ per region. The success criterion is that the central maximum stands above its two minima by 5.7 standard deviations of that scatter, so that the fringe is visible beyond the draw for any seed, with the total within a few percent of the real line's. (c) *The lattice's own number*. The law's line stepped in real arithmetic gives the screen's inflow peak at the interval 68 on the lattice, with the centroid at 73, against 63, with the centroid at 70, for a continuum wave in the same geometry; the click line resolves the five intervals (COMPUTED, CLICKS). Under the meeting's credit nothing changes here: light is one real line with nothing to pair, and the share is the record's own inflow. A world with one gap closed by the wall's own face is a closed slit by construction and not a which-way measurement, so it is not one of the gates [1, the rows against nature, (g)][3, the click is the meeting of the future with the past].

9.3 Bell: the world

In the model's words, entanglement is one event with two trees. The pair is laid as the pair family, two real lines of light never summed at a Node, nature's HH and VV, with equal parts, from one symmetric source, one beam to each side. Each side's detector is a region at least half a wavelength across that reports the two parts' signed level sums at its ports in each interval, the instrument's labelled reading, and knows nothing of angles; in this first design no device stands in the beam, and a device acting on each part separately, with the detectors still reporting the parts' signed sums, is the same credit with its coefficients relocated. The reader applies the declared setting, a pair of integers (p, q) , with the port coefficients $e(+) = (p, q)$ and $e(-) = (-q, p)$, exactly orthogonal, so each side's marginal is one half for any setting. The settings of the Clauser-Horne-Shimony-Holt test are the pairs (1, 0), (1, 1), (12, 5) and (5, 12), at 0, 45, 22.62 and 67.38 degrees, one setting pair per world, four worlds and one drawn pair each, with no cosine and no root in the implementation. The credit is the one form of the click (Section 5.2): the joint share pairs the parts by their label

across the two sides through the root,

$$J(p_A, p_B) = \left| \sum_k c_k e_k(p_A) \alpha_k(p_A) e_k(p_B) \beta_k(p_B) \right|^2, \quad c = (1, 1), \quad (17)$$

with α_k and β_k the part k 's amplitude at the port on each side and $e_k(p)$ the port's integer coefficient on the part k . Accumulated over the window on both members of the level pair, $J = \sum_{\text{window}} (\sum_k c_k \text{now}_k^A \text{now}_k^B)^2 + (\sum_k c_k \text{before}_k^A \text{before}_k^B)^2$ with $c_k = e_k(p_A) e_k(p_B)$, positive for a travelling record up to the window's ends, so a level sum passing through 0 gives no 0/0. The four densities over their sum give E as a rational number with the norms cancelling. The joint share is never the square of a local sum, which one detector could form alone and which gives $E = \sin 2a \sin 2b$ and S at most 2. One draw is made per pair, ordered in time: the first side from its marginal and the partner from the conditional. A comb with two gaps and a port shift cannot serve as the analyser, since a port of half a wavelength spans about 80 degrees of the comb's phase and cannot read it, which is the uncertainty relation of Section 5.3 (ASSUMPTION, CLICKS). The reader of the joint share belongs to the implementation's next version, with a dedicated test of the pairing through the root against the local sum, of the determinism of equal parts, and of the two local numbers as a lower fence [3, Bell in nature's own layout; the generic Node is closed for building].

9.4 Bell: what the model says it must give

Three layers, two from the algebra of the GameBoard and the third a declaration. (1) Rule3 gives each part's amplitudes at the ports: the two parts of one beam, laid equal, are two records of one family stepped by the same line along the same path, identical bit for bit at every Node and interval (THEOREM, GAMEBOARD). (2) The meeting with the linearity of Rule3 gives the joint amplitude: the realised click at A's port weights the parts by A's analyser, is read back to the root where the two parts were one event and forward to B, where B's analyser weights them, and the root sums the two branches, $\sum_k c_k e_k(a) e_k(b) = \cos(a - b)/\sqrt{2}$ for equal parts (DERIVED, CLICKS). (3) That the share is the density of the meeting, its square, is Born's rule for a pair [43]: nothing in Rule3 forces it, and it is the least non-local credit that reaches nature's correlations with every quantum counted (ASSUMPTION, FITTED; CLICKS). From the three, with the pair's common phase θ uniform over the pairs and A's + port at a and B's at b ,

$$\begin{aligned} \text{the meeting:} & \quad P(++) = \frac{1}{2} \cos^2(a - b), \quad E(a, b) = \cos 2(a - b), \quad S = 2\sqrt{2}; \\ \text{a local credit by the shares:} & \quad P(++) = \frac{1}{4} + \frac{1}{8} \cos 2(a - b), \quad E = \frac{1}{2} \cos 2(a - b), \quad S = \sqrt{2}; \\ \text{a local credit by the sign:} & \quad E = 1 - 4|a - b|/\pi \quad (|a - b| \leq \pi/2), \quad S = 2; \\ \text{the credit by the contrast:} & \quad E = \cos 2(a - b) \text{ at the efficiency } 2/\pi, \quad S = 2\sqrt{2}, \end{aligned} \quad (18)$$

The second line follows by averaging $\cos^2(a - \theta) \cos^2(b - \theta)$ over θ , the third from the band of width $|a - b|$ in each half-period where the two signs differ, and the fourth from the integral of $\cos 2(b - \theta)$ over the half-period where $\cos 2(a - \theta) > 0$, against the mean weight $(1/\pi) \int |\cos 2u| du$ (DERIVED, CLICKS). So $E(a, b) = \cos 2(a - b)$ under the meeting. For the declared pairs it is exact, Lagrange's identity of the reader on equal parts: $E = ((p_{APB} + q_{AQB})^2 - (p_{AQB} - q_{APB})^2) / ((p_A^2 + q_A^2)(p_B^2 + q_B^2))$, so at the four settings $E = 119/169, -119/169, 120/169$ and $120/169$ and $S = 478/169 = 2.8284$ exactly, with every pair credited on both sides, an efficiency of 1 by construction and the marginal one half on each side whatever the partner's setting (COMPUTED, CLICKS). For parts of unequal amplitudes, with $M_k = \alpha_k \beta_k$ and $r = M_2/M_1$, the same credit gives $E(a, b) = \cos 2a \cos 2b + \rho \sin 2a \sin 2b$

with $\rho = 2r/(1 + r^2)$, the marginal $P(A+) = (\cos^2 a + r^2 \sin^2 a)/(1 + r^2)$ independent of b , and $S = \sqrt{2}(1 + \rho)$ at the ideal angles and $(238 + 240\rho)/169$ at the integer pairs: $\rho = 1$ is the cosine, $\rho = 0$ the product form $238/169 = 1.408$ of the local shares, and the sign alone the sawtooth with $S = 2$; a shortfall of a run reads as $1 - \rho$, a reading of the two parts' amplitudes at the ports (DERIVED; ρ and r GAMEBOARD, E and S CLICKS). This is the two-photon correlation of a pair with the unequal weights $(1, r)$, quantum theory's own.

What the GameBoard gives without the declaration is the second and third lines of Eq. (18), two theorems about local credits on the same world (THEOREM, CLICKS). A credit of each side by the shares of its ports gives $S = \sqrt{2}$. A credit of each side by the sign of its larger port, the pair's phase being a shared classical variable, gives the sawtooth, whose sum is exactly 2 for every set of four settings within one period, Bell's bound. A two-port detector that credits by its contrast and counts nothing where its two ports' shares are nearly equal gives $E = \cos 2(a - b)$ exactly at the efficiency $(2/\pi)V$ for a visibility V : the uncounted pairs are those near the contrast's zero, which is the detection loophole of Bell tests [27], a local form that reaches nature's cosine only below the loophole's bound, $2/3$ at best [28], as $2/\pi$ does (DERIVED, CLICKS). Bell's value in this model therefore sits in the credit rule, which is a declaration, and the local theorems state what the GameBoard gives without it. Nature beside the derivation: $E = \cos 2(a - b)$, and $S = 2.42 \pm 0.20$ in the loophole-free test of 2015 that measured the sum [30]; the two photon tests of that year violated the Clauser-Horne-Eberhard form of the inequality [28] at detection efficiencies near 0.75 [31][32]. No signal passes between the sides: the detectors read locally, the root joins by linearity, and each side's marginal is one half. No joint credit is declared by an instrument, since two detectors are never at one Node, and no settings are chosen by the GameBoard's own past. A superdeterministic world, in which the settings are set by bodies sharing the source's history, would be a world of its own [33] and is not this one [1, the rows against nature, (h)][3, the click is the meeting of the future with the past; Bell in nature's own layout].

10 Constants, calibration and limits

10.1 The constants

The causal bound is one Link per interval, and light's own speed at long wavelength is $1/\sqrt{3}$ Links per interval (Section 3.3). The quantum of action is one click, whose energy is the quantum's norm T . The quantum of distance is the Link and that of time the interval; nothing between two Nodes or two intervals is observed. Newton's constant is not declared: a body whose wells are s quanta per interval makes the massless field's rest $3G(r)s$ at r Links over the field's divisor, $3s/(4\pi r)$ far away, and the level slows the clock by the potential $\Phi = -c/\Gamma$, so (DERIVED, GAMEBOARD)

$$G = \frac{3}{4\pi\Gamma E_g} \text{ per quantum of well, in Links and intervals,} \quad (19)$$

a reading of Γ , the file's E_g and the lattice. It is the clock's constant over c^2 : with $c^2 = 1/3$ in Links and intervals, $G_{\text{clock}} = 1/(4\pi\Gamma E_g)$ is the number per quantum of well that lowers the clock's share, and Kepler's G , read from the orbits of matter as clicks, is $[2 \cos \omega_0 / (1 + \cos \omega_0)] G_{\text{clock}}$ by Eq. (14), 0.8 of it at $[2, 3]$ and 1 as the gap closes (DERIVED, CLICKS). A world at a smaller Γ has stronger gravity per unit of content, and ratios between forms carry G once and cancel it. The vacuum content, the massless field's declared rest, is the model's cosmological constant, 0 by nature's clicks (Section 6.5) [1, the constants].

The free numbers. The file holds Γ , T , matter's numerator m , the binding field's pair, gravity's E_g and the vacuum content (Table 4). Γ is free and large against a proton's count, as the Link is

small against a proton, and no ratio fixes it yet. The pair (T, Γ) holds one physical number, $\sqrt{T}/\Gamma$: the level of one quantum of light is $\sqrt{T}/\sin\omega$, the pace reads levels, so light's action per quantum on a body's content is $\sqrt{T}/(\Gamma \sin\omega_L)$, 0.07 of Γ for one quantum per Node at a wavelength of 8 Links at $T = 2^{15}$ and $\Gamma = 6,000$. Nature's coupling per quantum is small, and what fixes $\sqrt{T}/\Gamma$ is open [1, no free number but the mass; what is open, item 9].

10.2 Constants the model derives with no free number

A constant of nature is derived as a dimensionless number, or as a dimensional one once the units are fixed from nature. Five candidates, each computed before its run [1, the rows against nature, (v)]:

- (1) *The bending factor of light against a matter clock*: 1.9 on the lattice in the first approximation, against general relativity's 2 (Section 8, (b)); its computation to three digits is the sharpest test the model offers (COMPUTED, CLICKS).
- (2) *The strength of gravity against the quantum*, $Gm^2/(\hbar c)$: under every divisor 1 the coupling of one quantum to another is of order one, against nature's 1.75×10^{-45} for the electron and 5.9×10^{-39} for the proton; the model takes it as the file's one number E_g , about 3×10^{42} for nature (COMPUTED under every divisor 1, ASSUMPTION with the binding field).
- (3) *The fine-structure constant*: the charge's field's well of one charged quantum against the content field's well of one quantum, both at the divisor 1, and α itself as light's coupling to the charge, still to compute against 1/137.036 (open).
- (4) *The mass ratio $m_p/m_e = 1,836$* : the ratio of a cloud's count to one quantum, an input and not derived, which fixes no Γ (Section 7.3).
- (5) *The lattice constants*: $3G(0) = 0.7582$, $3/(4\pi)$ far away and light's $1/\sqrt{3}$ Links per interval, numbers of the six Ports and not of nature (COMPUTED, GAMEBOARD).

10.3 Nature's numbers enter only as clicks

Every measured number of an experiment, whether a frequency, a wavelength, a deflection, a redshift or a coincidence rate, is first written as a click pattern on a declared detector against a declared clock, and only then set beside the model's click pattern. The families' pairs and the units Γ and T , what in nature is one Link and one interval, are fixed by that match and by nothing read off the GameBoard; the two ratios the model fixes without it are light's speed of $1/\sqrt{3}$ Links per interval and the causal bound of 1 (ASSUMPTION, CLICKS) [1, the reading; nature's numbers enter the board only as clicks][3, the experiments needed]. Read this way, a clock of nature is a click rate, a length is a click's place on a screen, and an energy or a mass is a count of quanta; a family's pair sets its rest rotation $\cos\omega_0 = \text{num}/\text{den}$, so one click rate of nature per family fixes its pair once Γ and T fix the unit. The calibration series, one number of the model against one click pattern of nature per item:

- (1) *The two speeds*. Gravity's own events and light at a content x arrive apart by $2xL\sqrt{3}$ intervals over L Links (Section 6.5); nature's GW170817 bounds gravity's lead over light at 7×10^{-16} of its speed [45], which fixes $c_{\text{vac}} = 0$. Beside it, gravity's divisor is bounded from below by light's own fringes: a light packet writes its own form into gravity's field over E_g , and where that well reached one level the packet's fringes would move with the intensity, which nature's fringes never do; so E_g is at least the packet's summed form over T for the strongest light measured, and the light-only worlds use 10^6 at $\Gamma = 6,000$ (DERIVED, CLICKS).
- (2) *Light's dispersion on the lattice*. Along an axis $\cos\omega = (2 + \cos k)/3$, so the group speed falls as $1 - 0.083k^2$, by 5 percent at a wavelength of 8 Links and 0.08 percent at 64. The photons of one gamma-ray burst, up to the TeV range, arrive together: a dependence of light's speed on

the square of its energy, which is the lattice's form, is bounded below about 3×10^{-16} at 1 TeV [47], and a linear dependence far more tightly [48]. So $k = 2\pi \text{ Link}/\lambda$ is below about 6×10^{-8} per Link at 1 TeV, and the Link is below about 10^{-26} metres, the first bound on the unit of Γ (COMPUTED, CLICKS).

- (3) *Matter's two slits.* A matter packet's band curvature near its bottom, $\omega \approx \omega_0 + k^2/(2m^*)$, gives $v = k/m^*$; the fringe spacing $\lambda L/d$ and a time of flight between two detectors give m^* in Links and intervals, and nature's $h/(mv)$ with the units of (1) and (2) gives the matter pair.
- (4) *The light clock.* A packet between two mirror faces L Links apart ticks with the period $2L\sqrt{3}$ intervals at one face's detector; in a declared static well of content x its period is $2L\sqrt{3}/(1-2x)$, so its rate falls by $2x$, the Link's pace read twice, and a screen beside it reads the bending $2x$: the ratio is 1. A clock that reads the content once is a bound body's rotation, $f(\omega_0)x$, so the ratio 2 between light's bending and a clock's redshift is read between light and a body, as Section 8 has it (DERIVED, CLICKS).

Only the first number is fixed today, the vacuum content, and two are bounded, gravity's divisor and the Link. Until the rest are named, run and read, the model's numbers are its own and not nature's, its families are the rule's bands and not nature's particles, and the implementation's two gates (Section 9) show that the implementation works and are not these experiments.

To be determined by experiment. Four numbers of the model wait on nature's clicks, each with its experiment named (EXPERIMENT, CLICKS): the units Γ and T , by one click rate and one fringe spacing of nature against the model's; gravity's divisor E_g , by the bending of a known mass read as clicks; the matter pair's gap, which the post-Newtonian tests bound below 2×10^{-5} (Section 8); and the wall of the sign holder's Link parts, by Ampère's force against Coulomb's, already nature's $1/c^2$ (Section 8, (f)).

10.4 Hypotheses under their own names

Every hypothesis stands outside the law under its own name until it passes the three tests or is adopted by decision, and each says what would decide it [1, the hypotheses under their own names].

Hypothesis 1 (the conversion, the weak force). A click that takes a quantum of one family and gives it as quanta of others, the rotations adding, is the whole weak interaction, one Node's act, no field and no massive vector, and no family's row declares anything for it.

Hypothesis 2 (Gamma is not constant). $\Gamma_{t+1} = \Gamma_t + 6$, the step the six Ports of a Node and no free number; the rule's physics depends on the ratios c/Γ and m/Γ alone, so every pair steps with Γ and the wall and the amplitude unit are derived from Γ_t each interval; a pixel of a fixed count falls under the rising edge and dissolves, the distinct bound counts grow with Γ , light does not drift while the clocks of pixels do, the rule's Hubble rate $H = (6/\Gamma) |d \ln \omega_b / d \ln (c/\Gamma)|$, and $\Gamma = 6t$ reads the universe's age in intervals. The time road's price: a stepping Γ is Dirac's road, G falling as $1/\text{age}$; nature bounds G 's drift today under 10^{-13} per year against 7×10^{-11} per year for a G proportional to $1/\text{age}$, and in the law a coupling that decays weakens the bodies' binding with it unless the binding is a field of its own, as the binding field makes it [1, Gamma is not constant].

Hypothesis 3 (the three distances are the rule's). That α , m_e/m_P and m_p/m_e are derived; the algebra is closed on one root, two named assumptions, one assumption of the universe and one hypothesis, with no number but gravity's E_g , and is not closed toward nature by name [1, the tree].

Hypothesis 4 (the hill holds the body). A body stands as a stable extended fixed point where a field read with the opposite sign, of reach shorter than the hollow's, is held beside the hollow: below

the hill's reach both are Coulomb-like and the hill forbids the collapse, between the two reaches the hollow binds, and the body's size lies between the reaches. Decided by a run of the generator with a third field.

Hypothesis 5 (the cooling). A body above its minimum sheds its excess as quanta of its own family, messages of matter, and radiates light by its write while it breathes, the shedding stopping at the stable body theorem's minimum for its count; whether a body settles at the minimum is a question for a run. The number of quanta a cloud gives to settle: for a Gaussian cloud twice too wide in its own massless well, 13 percent leave on a chain, 20 on a plane, 23 in a box (DERIVED, an expectation written before any run) [1, the number of quanta a cloud gives to settle].

Hypothesis 6 (the anti-body). An anti-body would be the same record with the sign of its count reversed. The count is the record's share, which has no sign to reverse, so the anti-body has no line in the law, and the antineutron waits for a line of the record that is distinct from the neutron's.

Hypothesis 7 (the second level's amplitude is a count of its own). The second level's amplitude quantised as a count of its own, a circular quantum the unit and a body's charge its number of circular quanta, so that $|q_p| = |q_e|$ would be derived where the law as written leaves a body's charge continuous (Section 5.5); no line of the law as written does it and Theorem 2 may not allow it.

Hypothesis 8 (the holder of the sign rotates the plane, with a phase on the Links). The holder of the sign acts on a plane record by rotating it by the angle of its level over Γ , three integer shears with no remainder, and its three Link parts, sourced by the Wronskian's current of its readers at the wall E_s den T , rotate the arrivals through the $\pm a$ Ports by $\mp\theta_a$. Under it like and unlike charges are told apart, the Lorentz force is the lattice's own identity and the force between parallel charges is Coulomb's over γ (Section 8, (f) and (g)); decided by the deflection of two charged bodies moving in parallel against 0.995 at $0.1c$ and 0.866 at $0.5c$.

Beside them, by name: a third form of gravity, the massless field's level as a count of resting clicks relaxed each interval, which would trade away the exact inverse or causality; and a controlled phase of order one between two records, which Theorem 2 excludes from the acts alone [1, the fork, decided].

10.5 Findings against nature, open questions and what is not claimed

Findings against nature. The model states where it differs from nature, as computations from its lines and not as clicks. The proton is a cloud at least the electron's Compton length across, where nature's proton is 460 times smaller (Section 7.3). The charged cloud is heavier than the neutral one, where nature's neutron is the heavier (Section 7.3). With the diagonal tension alone, the massless field's travelling events carry the breathing polarisation and the lattice-aligned + polarisation and no $\times$, where nature's gravitational waves carry + and $\times$ at every orientation; and light's band along an axis against a diagonal is anisotropic by about 1 percent at a wavelength of 8 Links, falling as the square of the Link over the wavelength, which bounds the Link from nature's isotropy [1, what is open, item 23]. The law has no current's read for real records, and gravity's readers are real records, so a neutral gyroscope precesses by 0 from a rotating body's motion, where Gravity Probe B measured 37 ± 7 milliarcseconds per year against general relativity's 39 [36]; closing this would need gravity's readers to carry a phase and a wall a quarter of den T , nature's 4 against the lattice's 3, neither of which the law has (DERIVED, CLICKS) [3, general relativity in the law; the current's read]. And against Kepler's potential the post-Newtonian parameters leave 1 at the rule's own matter pair, by 25 to 33 percent (Section 8).

Open. The sign of the electric force and the magnetic force, which wait on Hypothesis 8; the unit of charge (Hypothesis 7); the matter pair’s gap, which nature’s tests bound (Section 8); what fixes Γ and $\sqrt{T}/\Gamma$; the fine-structure constant and the mass ratio; the hierarchy of gravity and electricity, decided with a reservation (Section 6.4); the stability theorem’s Hessian beyond the dilation and the functional beyond first order; whether a breathing body cools; the two-mode transfer rate of the absorption line (Section 7.4); and a current’s read for real records, which the three tests do not admit [1, what is open].

Not claimed. Any agreement with nature from a run, since no run’s number stands here; that nature is so; the Lorentz group, which is no symmetry of the lattice; which meeting becomes the click; that the share is the square of the meeting as anything but a declaration; the fine-structure constant, the mass ratio and what fixes Γ ; spin 1/2; a body’s settling; and the magnetic force, which waits on a hypothesis. Every hypothesis of the paper is named as one.

11 The implementation

The implementation, Universe24 [49][2], is the law’s implementation and nothing else: a GameBoard of Nodes, every family’s NodeState at every Node, and one interval of acts in the law’s order, each a call of Rule3 on whole-board arrays of integers, the hardware’s 64-bit integers at a declared width of at most 63 bits and Python’s integers above it, with every neighbour read through a Port. It holds no number of physics, no formula, no family’s name and no flag; every physical value comes from the run’s files; no act of the interval takes a square root, and no act reads a Node to decide anything. It consists of six modules of under 400 lines each, for the Node’s acts, the share as a reading, the reports, the receding face’s growth, the GameBoard that walks the interval forward and back, and the files with their loader, on a core of Rule3, the Ports and the integer bound. Three structural tests hold the state, the width and the names: every Node to the NodeState of Section 3.1, a flat list of lines that the step walks knowing no family, no dimension and no name; the width to the loader’s one choice; and every word of the files and of the output to the loader’s key tables and the reports. The implementation reads the holder of the sign by the family’s dimension and no coefficient from a Node’s own state, so the reversibility gate passes without exception. The next versions, named by the decisions, carry the pair family with the meeting’s joint credit and its reader, the two gates with their expectation files, and the rotation of Hypothesis 8 [2, the engine; the NodeState; how to add a family][3, the generic Node is closed for building].

The files. *The universe file* declares the integers (Γ , T and the width) and the families, each with its name, its pair and its shape, a family of quanta its dimension and a field its sources with its divisor and, for the massless field, its rest; the rule derives the rest. *The world file* declares the GameBoard’s shape and faces, the intervals, the bodies by their family with their Nodes and counts, the messages by their family, axis, wave number, amplitude, envelope and phase, and the detectors by their regions; a missing key is refused by name, and no default is written. *The mode file* beside the world holds the generator’s levels of every body and message. *The expectation file* holds the prediction written before the run: for a screen the regions, the window, the expected counts and the draw’s seed, and for Bell the ports, the settings and the expected E and S [2, the loader and the files; how to run a world].

A run. The generator lays the bodies and the messages; the check at the start compares each body’s declared count with its record’s share; the run is headless and writes one output file per world. The output holds the verdict, LAWFUL or REFUSED with the reason; one click line per family, region

and interval with the net current through the region's front boundary, which is the measurement; a field line per region with a family's density over it, which is a diagnostic; and diagnostic totals. A reader then credits a screen's clicks against the expectation file, by the largest remainders and by one draw at the declared seed. A world is run for results only after the reversibility gate says MATCH on it [2, the output; how to run a world].

How to use the simulator, in a few lines. Write a world file with the GameBoard's shape and faces, the universe file it names, the bodies, the messages and the detectors; every number of physics is in the files and none in the code. Lay it with `PYTHONPATH=src python tools/pixel.mode.py --input world.json`, which writes the mode file with every body's fixed point and every message's levels. Run it headless with `PYTHONPATH=src python tools/run_inputs.py --out runs/first world.json`, which writes the verdict, the click lines and the field lines. Check it with `PYTHONPATH=src python tools/back_in_time.py --intervals 40 world.json`, which runs the world forward and back and says MATCH or MISS. Read it with `tools/click_counts.py`, which credits a screen's clicks against the expectation file; `tools/look/record.py` with `tools/look/page.py` draws the GameBoard as a diagnostic.

Where each section lives. The line is in `core/rule3.py` and the Ports in `core/ports.py` (Section 3); the paces and the guard in `features/read`; the readings of the form, the Wronskian, the currents and the tension in `node.py` and the share in `share.py` (Sections 4 and 5); the one write in `features/hold` and `features/write` and the start in `features/start`; the detectors' reports in `reports.py` and the receding face in `growth.py`; the interval and its inverse in `game_board.py`; the families from the rule in `loader/derived.py` (Section 6); the generator in `tools/pixel.mode.py` (Section 7); and the two gates' worlds under `examples/events/` (Section 9) [2, how to run a world].

The tests. Every change runs lint, formatting, strict types and the tests of the changed files and their consumers. The tests hold the rule's arithmetic to one function, every shift across a Link to the Ports, every integer literal of the code to the law's own, every module to integers with no root, and the Node to its state and acts; they check the interval on a closed cube under the 48 symmetries with the reversibility gate, the share's identity against the currents, the tension as the conservation of the current, the birth of light by the write, the receding face, the rest of every pair as its line's own fixed point, and the generator against real arithmetic [2, the gates].

To test a hypothesis. Its line is written in the law's document first, under its own name, with what would decide it. The expected number is computed from the implementation's own lines and written down before the run. The hypothesis is built as one pure function of arrays that calls Rule3 alone, with its own test. The run's reading is compared with the written number: a difference is a defect of the implementation, never a finding, and a number outside its band with no defect found names the missing law as a hypothesis. No expectation is rewritten after a run, and no body's number is tuned to a result [50, the blind number first].

12 Conclusion

One line of integer arithmetic at every Node of a cubic lattice, with one remainder kept at the Node, is reversible to the bit, conserves a quadratic form with its current, and carries the momentum flux as the wave's own stress. On that lattice there are only events that split; the future and the past are the same split events run forward from an emitter and read backward from a click, and a click is their meeting at a declared detector, with the choice among the meetings as the one non-local element. Families are bands of the rule, each one row of a file, and the universe holds as few free

numbers as the rule allows. Bodies are the rule’s fixed points, the extended ones behaving as nature’s bodies and the compact ones as collapsed matter; a cluster of bodies is the same structure one level up, bound by the three fields by rank, with the click the only thing between its members, so that an observer, a cluster itself, sees only clicks. Gravity is a level that slows a Node’s clock once and its Links twice. In the model’s own potential two of its features were fitted, one to light’s bending and one to the perihelion; against the potential that Kepler’s orbits measure, the same forms reach Einstein’s only as the matter family’s gap closes, which bounds that gap and stands against the rule’s own matter pair. Bell’s value rests on a credit rule chosen to reach nature’s correlations. Each such result is marked as a consistency check and not a prediction. Lorentz’s factor at a body’s own light speed and the universality of free fall for extended bodies follow without fitting; the magnetic force between moving charges does not follow from the rule as it stands, and the paper names the hypothesis that would give it. Every formula is fenced as a reading of the lattice or as a formula of clicks, and the formulas of clicks stand with their statuses as the model’s list of what it gives. The two slits and Bell are stated as the implementation’s two gates, what each must give; the next step is to run them as gates, to compare a body’s clicks with nature’s, and to fix the model’s remaining numbers by nature’s clicks one at a time. We hold the whole to be the least computation that shows nature’s clicks, with the least state, the least non-locality and the least randomness; why nature is so, we do not claim to know.

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Data and code availability. The law’s document, the implementation and the figures’ scripts are archived at Zenodo under the concept DOI 10.5281/zenodo.22738746 [49], under the MIT licence. The paper cites no run, and the worlds of its two experiments are in the archive as files.

Author contributions. The author laid down every postulate and decision of the model, directed the work and wrote the paper.

The use of a large language model. The author used a large language model (Claude, Anthropic, through Claude Code, 2026) extensively as a tool, under the author’s direction, for the software, the computations, the mathematical derivations and their checks, and the drafting of the text. The model is not an author. The author reviewed every statement and takes full responsibility for the content. No figure was made by a generative tool; every figure is drawn by a script from the definitions or from the law’s document.

References

- [1] The law of the GameBoard, `docs/ALGEBRA.md` of the archive [49], cited by the name of its section: the GameBoard and the families (the objects, a family’s declaration, the interval, readings and measurements); Rule3 (the line, the direction, the four acts, the paces, the bound, the self-source, the conserved form); the click (the booking, the count is the record’s share, the click is the detector’s report, no click names a Node, the algebra of the exclusivity); the primitives; the stable body (what a body is, the well, the functional and the dilation, the surplus leaves, the generator, the velocity); what the law says of nature

(the reading, with no waves on the event board, what produces a click is not known and nature's numbers enter the board only as clicks; the hypotheses under their own names; the postulates; the constants; the rows against nature); what is open.

- [2] The engine, `docs/ENGINE.md` of the archive [49]: the words, the NodeState, the interval, the loader and the files, the output, how to run a world, how to add a family, the gates.
- [3] The decisions, `docs/HIGHLIGHTS.md` of the archive [49]: the model's decisions in force, one line each, cited by the line's name.
- [4] K. Zuse, *Rechnender Raum* (Vieweg, Braunschweig, 1969); English translation: *Calculating Space*, MIT Technical Translation AZT-70-164-GEMIT (MIT Project MAC, Cambridge, 1970).
- [5] E. Fredkin and T. Toffoli, Conservative logic, *International Journal of Theoretical Physics* 21, 219 (1982), <https://doi.org/10.1007/BF01857727>.
- [6] N. Margolus, Physics-like models of computation, *Physica D* 10, 81 (1984), [https://doi.org/10.1016/0167-2789\(84\)90252-5](https://doi.org/10.1016/0167-2789(84)90252-5).
- [7] T. Toffoli and N. Margolus, Invertible cellular automata: a review, *Physica D* 45, 229 (1990), [https://doi.org/10.1016/0167-2789\(90\)90185-R](https://doi.org/10.1016/0167-2789(90)90185-R).
- [8] S. Wolfram, *A New Kind of Science* (Wolfram Media, Champaign, 2002).
- [9] G. 't Hooft, Duality between a deterministic cellular automaton and a bosonic quantum field theory in 1+1 dimensions, *Foundations of Physics* 43, 597 (2013), <https://doi.org/10.1007/s10701-013-9702-z>.
- [10] G. 't Hooft, *The Cellular Automaton Interpretation of Quantum Mechanics*, *Fundamental Theories of Physics* 185 (Springer, Cham, 2016), <https://doi.org/10.1007/978-3-319-41285-6>.
- [11] R. P. Feynman, Simulating physics with computers, *International Journal of Theoretical Physics* 21, 467 (1982), <https://doi.org/10.1007/BF02650179>.
- [12] R. Courant, K. Friedrichs and H. Lewy, Über die partiellen Differenzengleichungen der mathematischen Physik, *Mathematische Annalen* 100, 32 (1928), <https://doi.org/10.1007/BF01448839>.
- [13] K. G. Wilson, Confinement of quarks, *Physical Review D* 10, 2445 (1974), <https://doi.org/10.1103/PhysRevD.10.2445>.
- [14] Y. Aharonov, L. Davidovich and N. Zagury, Quantum random walks, *Physical Review A* 48, 1687 (1993), <https://doi.org/10.1103/PhysRevA.48.1687>.
- [15] I. Bialynicki-Birula, Weyl, Dirac, and Maxwell equations on a lattice as unitary cellular automata, *Physical Review D* 49, 6920 (1994), <https://doi.org/10.1103/PhysRevD.49.6920>.
- [16] D. A. Meyer, From quantum cellular automata to quantum lattice gases, *Journal of Statistical Physics* 85, 551 (1996), <https://doi.org/10.1007/BF02199356>.
- [17] J. A. Wheeler and R. P. Feynman, Interaction with the absorber as the mechanism of radiation, *Reviews of Modern Physics* 17, 157 (1945), <https://doi.org/10.1103/RevModPhys.17.157>.
- [18] J. G. Cramer, The transactional interpretation of quantum mechanics, *Reviews of Modern Physics* 58, 647 (1986), <https://doi.org/10.1103/RevModPhys.58.647>.
- [19] Y. Aharonov, P. G. Bergmann and J. L. Lebowitz, Time symmetry in the quantum process of measurement, *Physical Review* 134, B1410 (1964), <https://doi.org/10.1103/PhysRev.134.B1410>.
- [20] Y. Aharonov and L. Vaidman, The two-state vector formalism: an updated review, *Lecture Notes in Physics* 734, 399 (Springer, Berlin, 2008), https://doi.org/10.1007/978-3-540-73473-4_13.
- [21] H. Price, *Time's Arrow and Archimedes' Point: New Directions for the Physics of Time* (Oxford University Press, New York, 1996).
- [22] K. B. Wharton and N. Argaman, Colloquium: Bell's theorem and locally mediated reformulations of quantum mechanics, *Reviews of Modern Physics* 92, 021002 (2020), <https://doi.org/10.1103/RevModPhys.92.021002>.

- [23] N. Harrigan and R. W. Spekkens, Einstein, incompleteness, and the epistemic view of quantum states, *Foundations of Physics* 40, 125 (2010), <https://doi.org/10.1007/s10701-009-9347-0>.
- [24] J. S. Bell, On the Einstein Podolsky Rosen paradox, *Physics Physique Fizika* 1, 195 (1964), <https://doi.org/10.1103/PhysicsPhysiqueFizika.1.195>.
- [25] J. F. Clauser, M. A. Horne, A. Shimony and R. A. Holt, Proposed experiment to test local hidden-variable theories, *Physical Review Letters* 23, 880 (1969), <https://doi.org/10.1103/PhysRevLett.23.880>.
- [26] A. Fine, Hidden variables, joint probability, and the Bell inequalities, *Physical Review Letters* 48, 291 (1982), <https://doi.org/10.1103/PhysRevLett.48.291>.
- [27] P. M. Pearle, Hidden-variable example based upon data rejection, *Physical Review D* 2, 1418 (1970), <https://doi.org/10.1103/PhysRevD.2.1418>.
- [28] P. H. Eberhard, Background level and counter efficiencies required for a loophole-free Einstein-Podolsky-Rosen experiment, *Physical Review A* 47, R747 (1993), <https://doi.org/10.1103/PhysRevA.47.R747>.
- [29] A. Garg and N. D. Mermin, Detector inefficiencies in the Einstein-Podolsky-Rosen experiment, *Physical Review D* 35, 3831 (1987), <https://doi.org/10.1103/PhysRevD.35.3831>.
- [30] B. Hensen et al., Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres, *Nature* 526, 682 (2015), <https://doi.org/10.1038/nature15759>.
- [31] M. Giustina et al., Significant-loophole-free test of Bell's theorem with entangled photons, *Physical Review Letters* 115, 250401 (2015), <https://doi.org/10.1103/PhysRevLett.115.250401>.
- [32] L. K. Shalm et al., Strong loophole-free test of local realism, *Physical Review Letters* 115, 250402 (2015), <https://doi.org/10.1103/PhysRevLett.115.250402>.
- [33] S. Hossenfelder and T. Palmer, Rethinking superdeterminism, *Frontiers in Physics* 8, 139 (2020), <https://doi.org/10.3389/fphy.2020.00139>.
- [34] T. Regge, General relativity without coordinates, *Il Nuovo Cimento* 19, 558 (1961), <https://doi.org/10.1007/BF02733251>.
- [35] L. Bombelli, J. Lee, D. Meyer and R. D. Sorkin, Space-time as a causal set, *Physical Review Letters* 59, 521 (1987), <https://doi.org/10.1103/PhysRevLett.59.521>.
- [36] C. M. Will, The confrontation between general relativity and experiment, *Living Reviews in Relativity* 17, 4 (2014), <https://doi.org/10.12942/lrr-2014-4>.
- [37] P. Touboul et al., MICROSCOPE mission: final results of the test of the equivalence principle, *Physical Review Letters* 129, 121102 (2022), <https://doi.org/10.1103/PhysRevLett.129.121102>.
- [38] S. I. Pekar, *Untersuchungen über die Elektronentheorie der Kristalle* (Akademie-Verlag, Berlin, 1954).
- [39] G. H. Derrick, Comments on nonlinear wave equations as models for elementary particles, *Journal of Mathematical Physics* 5, 1252 (1964), <https://doi.org/10.1063/1.1704233>.
- [40] S. I. Pohozaev, Eigenfunctions of the equation $\Delta u + \lambda f(u) = 0$, *Soviet Mathematics Doklady* 6, 1408 (1965).
- [41] N. G. Vakhitov and A. A. Kolokolov, Stationary solutions of the wave equation in a medium with nonlinearity saturation, *Radiophysics and Quantum Electronics* 16, 783 (1973), <https://doi.org/10.1007/BF01031343>.
- [42] H. Yukawa, On the interaction of elementary particles. I, *Proceedings of the Physico-Mathematical Society of Japan* 17, 48 (1935), https://doi.org/10.11429/ppmsj1919.17.0_48.
- [43] M. Born, Zur Quantenmechanik der Stoßvorgänge, *Zeitschrift für Physik* 37, 863 (1926), <https://doi.org/10.1007/BF01397477>.
- [44] G. Bressi, G. Carugno, F. Della Valle, G. Galeazzi, G. Ruoso and G. Sartori, Testing the neutrality of matter by acoustic means in a spherical resonator, *Physical Review A* 83, 052101 (2011), <https://doi.org/10.1103/PhysRevA.83.052101>.

- [45] B. P. Abbott et al. (LIGO Scientific Collaboration, Virgo Collaboration, Fermi Gamma-ray Burst Monitor and INTEGRAL), Gravitational waves and gamma-rays from a binary neutron star merger: GW170817 and GRB 170817A, *The Astrophysical Journal Letters* 848, L13 (2017), <https://doi.org/10.3847/2041-8213/aa920c>.
- [46] I. I. Shapiro, Fourth test of general relativity, *Physical Review Letters* 13, 789 (1964), <https://doi.org/10.1103/PhysRevLett.13.789>.
- [47] V. A. Acciari et al. (MAGIC Collaboration), Bounds on Lorentz invariance violation from MAGIC observation of GRB 190114C, *Physical Review Letters* 125, 021301 (2020), <https://doi.org/10.1103/PhysRevLett.125.021301>.
- [48] A. A. Abdo et al. (Fermi GBM and LAT Collaborations), A limit on the variation of the speed of light arising from quantum gravity effects, *Nature* 462, 331 (2009), <https://doi.org/10.1038/nature08574>.
- [49] A. Gonen, Universe24: a discrete simulator of local physical laws (Reality Theory), Zenodo (2026), <https://doi.org/10.5281/zenodo.22738746> (the concept DOI, resolving to the latest version).
- [50] The shared way of working, `skills/workflow.md` of the archive [49]: the three tests, the six verbs, the blind number first, the short procedure.